RF Microelectronics

Second Edition



Behzad Razavi

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RF MICROELECTRONICS

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Behzad Razavi

University of California, Los Angeles



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To the memory of my parents

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PREFACE TO THE SECOND EDITION

In the 14 years since the first edition of this book, RF IC design has experienced a dramatic metamorphosis. Innovations in transceiver architectures, circuit topologies, and device structures have led to highly-integrated "radios" that span a broad spectrum of applications. Moreover, new analytical and modeling techniques have considerably improved our understanding of RF circuits and their underlying principles. A new edition was therefore due.

The second edition differs from the first in several respects:

1. I realized at the outset—three-and-a-half years ago—that simply adding "patches" to the first edition would not reflect today's RF microelectronics. I thus closed the first edition and began with a clean slate. The two editions have about 10% overlap.
2. I wanted the second edition to contain greater pedagogy, helping the reader understand both the fundamentals and the subtleties. I have thus incorporated hundreds of examples and problems.
3. I also wanted to teach *design* in addition to analysis. I have thus included step-by-step design procedures and examples. Furthermore, I have dedicated Chapter 13 to the step-by-step transistor-level design of a dual-band WiFi transceiver.
4. With the tremendous advances in RF design, some of the chapters have inevitably become longer and some have been split into two or more chapters. As a result, the second edition is nearly three times as long as the first.

Suggestions for Instructors and Students

The material in this book is much more than can be covered in one quarter or semester. The following is a possible sequence of the chapters that can be taught in one term with reasonable depth. Depending on the students' background and the instructor's preference, other combinations of topics can also be covered in one quarter or semester.

Chapter 1: Introduction to RF and Wireless Technology

This chapter provides the big picture and should be covered in about half an hour.

Chapter 2: Basic Concepts in RF Design

The following sections should be covered: General Considerations, Effects of Nonlinearity (the section on AM/PM Conversion can be skipped), Noise, and Sensitivity and Dynamic Range. (The sections on Passive Impedance Transformation, Scattering Parameters, and Analysis of Nonlinear Dynamic Systems can be skipped.) This chapter takes about six hours of lecture.

Chapter 3: Communication Concepts

This chapter can be covered minimally in a quarter system—for example, Analog Modulation, Quadrature Modulation, GMSK Modulation, Multiple Access Techniques, and the IEEE802.11a/b/g Standard. In a semester system, the concept of signal constellations can be introduced and a few more modulation schemes and wireless standards can be taught. This chapter takes about two hours in a quarter system and three hours in a semester system.

Chapter 4: Transceiver Architectures

This chapter is relatively long and should be taught selectively. The following sections should be covered: General Considerations, Basic and Modern Heterodyne Receivers, Direct-Conversion Receivers, Image-Reject Receivers, and Direct-Conversion Transmitters. In a semester system, Low-IF Receivers and Heterodyne Transmitters can be covered as well. This chapter takes about eight hours in a quarter system and ten hours in a semester system.

Chapter 5: Low-Noise Amplifiers

The following sections should be covered: General Considerations, Problem of Input Matching, and LNA Topologies. A semester system can also include Gain Switching and Band Switching or High-IP_2 LNAs. This chapter takes about six hours in a quarter system and eight hours in a semester system.

Chapter 6: Mixers

The following sections should be covered: General Considerations, Passive Downconversion Mixers (the computation of noise and input impedance of voltage-driven sampling mixers can be skipped), Active Downconversion Mixers, and Active Mixers with High IP_2. In a semester system, Active Mixers with Enhanced Transconductance, Active Mixers with Low Flicker Noise, and Upconversion Mixers can also be covered. This chapter takes about eight hours in a quarter system and ten hours in a semester system.

Chapter 7: Passive Devices

This chapter may not fit in a quarter system. In a semester system, about three hours can be spent on basic inductor structures and loss mechanisms and MOS varactors.

Chapter 8: Oscillators

This is a long chapter and should be taught selectively. The following sections should be covered: Basic Principles, Cross-Coupled Oscillator, Voltage-Controlled

Oscillators, Low-Noise VCOs. In a quarter system, there is little time to cover phase noise. In a semester system, both approaches to phase noise analysis can be taught. This chapter takes about six hours in a quarter system and eight hours in a semester system.

Chapter 9: Phase-Locked Loops

This chapter forms the foundation for synthesizers. In fact, if taught carefully, this chapter naturally teaches integer-N synthesizers, allowing a quarter system to skip the next chapter. The following sections should be covered: Basic Concepts, Type-I PLLs, Type-II PLLs, and PFD/CP Nonidealities. A semester system can also include Phase Noise in PLLs and Design Procedure. This chapter takes about four hours in a quarter system and six hours in a semester system.

Chapter 10: Integer-N Synthesizers

This chapter is likely sacrificed in a quarter system. A semester system can spend about four hours on Spur Reduction Techniques and Divider Design.

Chapter 11: Fractional-N Synthesizers

This chapter is likely sacrificed in a quarter system. A semester system can spend about four hours on Randomization and Noise Shaping. The remaining sections may be skipped.

Chapter 12: Power Amplifiers

This is a long chapter and, unfortunately, is often sacrificed for other chapters. If coverage is desired, the following sections may be taught: General Considerations, Classification of Power Amplifiers, High-Efficiency Power Amplifiers, Cascode Output Stages, and Basic Linearization Techniques. These topics take about four hours of lecture. Another four hours can be spent on Doherty Power Amplifier, Polar Modulation, and Outphasing.

Chapter 13: Transceiver Design Example

This chapter provides a step-by-step design of a dual-band transceiver. It is possible to skip the state-of-the-art examples in Chapters 5, 6, and 8 to allow some time for this chapter. The system-level derivations may still need to be skipped. The RX, TX, and synthesizer transistor-level designs can be covered in about four hours.

A solutions manual is available for instructors via the Pearson Higher Education Instructor Resource Center web site: pearsonhighered.com/irc; and a set of Powerpoint slides is available for instructors at **informit.com/razavi**. Additional problems will be posted on the book's website (**informit.com/razavi**).

—Behzad Razavi
July 2011

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PREFACE TO THE FIRST EDITION

The annual worldwide sales of cellular phones has exceeded $2.5B. With 4.5 million customers, home satellite networks comprise a $2.5B industry. The global positioning system is expected to become a $5B market by the year 2000. In Europe, the sales of equipment and services for mobile communications will reach $30B by 1998. The statistics are overwhelming.

The radio frequency (RF) and wireless market has suddenly expanded to unimaginable dimensions. Devices such as pagers, cellular and cordless phones, cable modems, and RF identification tags are rapidly penetrating all aspects of our lives, evolving from luxury items to indispensable tools. Semiconductor and system companies, small and large, analog and digital, have seen the statistics and are striving to capture their own market share by introducing various RF products.

RF design is unique in that it draws upon many disciplines unrelated to integrated circuits (ICs). The RF knowledge base has grown for almost a century, creating a seemingly endless body of literature for the novice.

This book deals with the analysis and design of RF integrated circuits and systems. Providing a systematic treatment of RF electronics in a tutorial language, the book begins with the necessary background knowledge from microwave and communication theory and leads the reader to the design of RF transceivers and circuits. The text emphasizes both architecture and circuit level issues with respect to monolithic implementation in VLSI technologies. The primary focus is on bipolar and CMOS design, but most of the concepts can be applied to other technologies as well. The reader is assumed to have a basic understanding of analog IC design and the theory of signals and systems.

The book consists of nine chapters. Chapter 1 gives a general introduction, posing questions and providing motivation for subsequent chapters. Chapter 2 describes basic concepts in RF and microwave design, emphasizing the effects of nonlinearity and noise.

Chapters 3 and 4 take the reader to the communication system level, giving an overview of modulation, detection, multiple access techniques, and wireless standards. While initially appearing to be unnecessary, this material is in fact essential to the concurrent design of RF circuits and systems.

Chapter 5 deals with transceiver architectures, presenting various receiver and transmitter topologies along with their merits and drawbacks. This chapter also includes a number of case studies that exemplify the approaches taken in actual RF products.

Chapters 6 through 9 address the design of RF building blocks: low-noise amplifiers and mixers, oscillators, frequency synthesizers, and power amplifiers, with particular attention to minimizing the number of off-chip components. An important goal of these chapters is to demonstrate how the system requirements define the parameters of the circuits and how the performance of each circuit impacts that of the overall transceiver.

I have taught approximately 80% of the material in this book in a 4-unit graduate course at UCLA. Chapters 3, 4, 8, and 9 had to be shortened in a ten-week quarter, but in a semester system they can be covered more thoroughly.

Much of my RF design knowledge comes from interactions with colleagues. Helen Kim, Ting-Ping Liu, and Dan Avidor of Bell Laboratories, and David Su and Andrew Gzegorek of Hewlett-Packard Laboratories have contributed to the material in this book in many ways. The text was also reviewed by a number of experts: Stefan Heinen (Siemens), Bart Jansen (Hewlett-Packard), Ting-Ping Liu (Bell Labs), John Long (University of Toronto), Tadao Nakagawa (NTT), Gitty Nasserbakht (Texas Instruments), Ted Rappaport (Virginia Tech), Tirdad Sowlati (Gennum), Trudy Stetzler (Bell Labs), David Su (Hewlett-Packard), and Rick Wesel (UCLA). In addition, a number of UCLA students, including Farbod Behbahani, Hooman Darabi, John Leete, and Jacob Rael, "test drove" various chapters and provided useful feedback. I am indebted to all of the above for their kind assistance.

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—Behzad Razavi
July 1997

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Even after several rounds of self-editing, it is possible that typos or subtle mistakes have eluded the author. Sometimes, an explanation that is clear to the author may not be so to the reader. And, occasionally, the author may have missed a point or a recent development. A detailed review of the book by others thus becomes necessary. The following individuals meticulously reviewed various chapters, discovered my mistakes, and made valuable suggestions:

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As with my other books, my wife, Angelina, typed the entire second edition in Latex and selflessly helped me in this three-and-a-half-year endeavor. I am grateful to her.

—Behzad Razavi

ABOUT THE AUTHOR

Behzad Razavi received the BSEE degree from Sharif University of Technology in 1985 and MSEE and PhDEE degrees from Stanford University in 1988 and 1992, respectively. He was with AT&T Bell Laboratories and Hewlett-Packard Laboratories until 1996. Since 1996, he has been associate professor and, subsequently, professor of electrical engineering at University of California, Los Angeles. His current research includes wireless transceivers, frequency synthesizers, phase-locking and clock recovery for high-speed data communications, and data converters.

Professor Razavi was an adjunct professor at Princeton University from 1992 to 1994, and at Stanford University in 1995. He served on the Technical Program Committees of the International Solid-State Circuits Conference (ISSCC) from 1993 to 2002 and VLSI Circuits Symposium from 1998 to 2002. He has also served as guest editor and associate editor of the *IEEE Journal of Solid-State Circuits, IEEE Transactions on Circuits and Systems,* and *International Journal of High Speed Electronics*.

Professor Razavi received the Beatrice Winner Award for Editorial Excellence at the 1994 ISSCC; the best paper award at the 1994 European Solid-State Circuits Conference; the best panel award at the 1995 and 1997 ISSCC; the TRW Innovative Teaching Award in 1997; the best paper award at the IEEE Custom Integrated Circuits Conference (CICC) in 1998; and McGraw-Hill First Edition of the Year Award in 2001. He was the co-recipient of both the Jack Kilby Outstanding Student Paper Award and the Beatrice Winner Award for Editorial Excellence at the 2001 ISSCC. He received the Lockheed Martin Excellence in Teaching Award in 2006; the UCLA Faculty Senate Teaching Award in 2007; and the CICC Best Invited Paper Award in 2009. He was also recognized as one of the top ten authors in the fifty-year history of ISSCC. He received the IEEE Donald Pederson Award in Solid-State Circuits in 2012.

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Optical Communications, and *Fundamentals of Microelectronics* (translated to Korean and Portuguese), and the editor of *Monolithic Phase-Locked Loops and Clock Recovery Circuits* and *Phase-Locking in High-Performance Systems*.

CHAPTER

1

INTRODUCTION TO RF AND WIRELESS TECHNOLOGY

Compare two RF transceivers designed for cell phones:

> *"A 2.7-V GSM RF Transceiver IC" [1] (published in 1997)*
>
> *"A Single-Chip 10-Band WCDMA/HSDPA 4-Band GSM/EDGE SAW-Less CMOS Receiver with DigRF 3G Interface and +90-dBm* IIP_2*" [2] (published in 2009)*

Why is the latter much more complex than the former? Does the latter have a higher performance or only greater functionality? Which one costs more? Which one consumes a higher power? What do all the acronyms GSM, WCDMA, HSDPA, EDGE, SAW, and IIP_2 mean? Why do we care?

The field of RF communication has grown rapidly over the past two decades, reaching far into our lives and livelihood. Our cell phones serve as an encyclopedia, a shopping terminus, a GPS guide, a weather monitor, and a telephone—all thanks to their wireless communication devices. We can now measure a patient's brain or heart activity and transmit the results wirelessly, allowing the patient to move around untethered. We use RF devices to track merchandise, pets, cattle, children, and convicts.

1.1 A WIRELESS WORLD

Wireless communication has become almost as ubiquitous as electricity; our refrigerators and ovens may not have a wireless device at this time, but it is envisioned that our homes will eventually incorporate a wireless network that controls every device and appliance. High-speed wireless links will allow seamless connections among our laptops, digital cameras, camcorders, cell phones, printers, TVs, microwave ovens, etc. Today's WiFi and Bluetooth connections are simple examples of such links.

How did wireless communication take over the world? A confluence of factors has contributed to this explosive growth. The principal reason for the popularity of wireless

communication is the ever-decreasing cost of electronics. Today's cell phones cost about the same as those a decade ago but they offer many more functions and features: many frequency bands and communication modes, WiFi, Bluetooth, GPS, computing, storage, a digital camera, and a user-friendly interface. This affordability finds its roots in *integration*, i.e., how much functionality can be placed on a single chip—or, rather, how few components are left off-chip. The integration, in turn, owes its steady rise to (1) the scaling of VLSI processes, particularly, CMOS technology, and (2) innovations in RF architectures, circuits, and devices.

Along with higher integration levels, the performance of RF circuits has also improved. For example, the power consumption necessary for a given function has decreased and the speed of RF circuits has increased. Figure 1.1 illustrates some of the trends in RF integrated circuits (ICs) and technology for the past two decades. The minimum feature size of CMOS

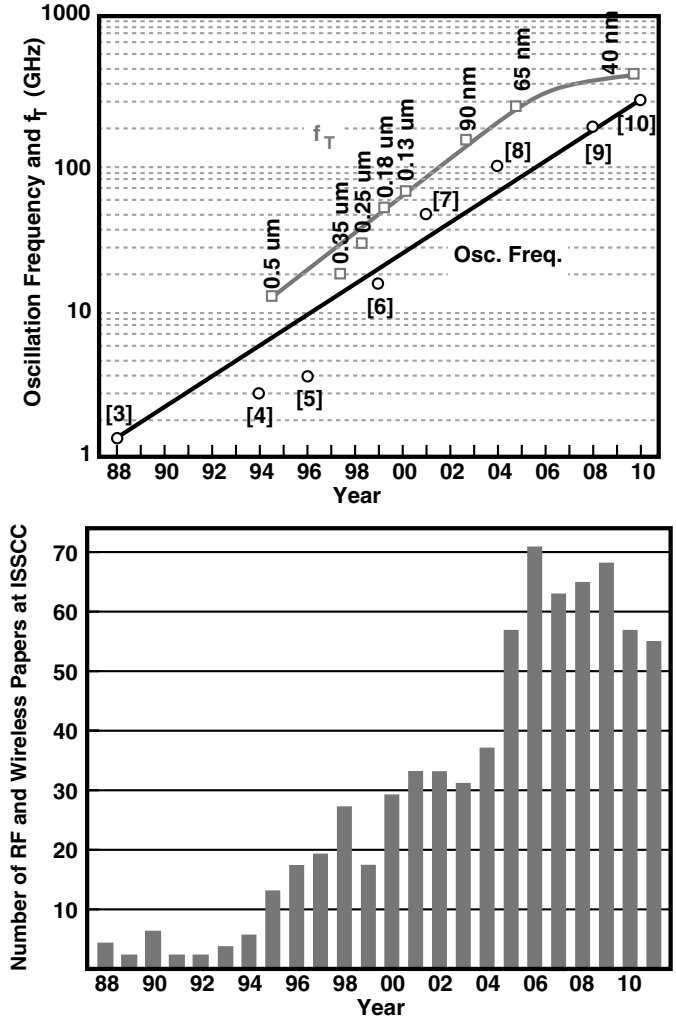


Figure 1.1 *Trends in RF circuits and technology.*

technology has fallen from 0.5 μm to 40 nm, the transit frequency,[1] f_T, of NMOS devices has risen from about 12 GHz to several hundred gigahertz, and the speed of RF oscillators has gone from 1.2 GHz to 300 GHz. Also shown is the number of RF and wireless design papers presented at the International Solid-State Circuits Conference (ISSCC) each year, revealing the fast-growing activity in this field.

1.2 RF DESIGN IS CHALLENGING

Despite many decades of work on RF and microwave theory and two decades of research on RF ICs, the design and implementation of RF circuits and transceivers remain challenging. This is for three reasons. First, as shown in Fig. 1.2, RF design draws upon a multitude of disciplines, requiring a good understanding of fields that are seemingly irrelevant to integrated circuits. Most of these fields have been under study for more than half a century, presenting a massive body of knowledge to a person entering RF IC design. One objective of this book is to provide the necessary background from these disciplines without overwhelming the reader.

Second, RF circuits and transceivers must deal with numerous trade-offs, summarized in the "RF design hexagon" of Fig. 1.3. For example, to lower the noise of a front-end amplifier, we must consume a greater power or sacrifice linearity. We will encounter these trade-offs throughout this book.

Third, the demand for higher performance, lower cost, and greater functionality continues to present new challenges. The early RF IC design work in the 1990s strove to integrate *one* transceiver—perhaps along with the digital baseband processor—on a single chip. Today's efforts, on the other hand, aim to accommodate multiple transceivers operating in different frequency bands for different wireless standards (e.g., Bluetooth, WiFi, GPS, etc.). The two papers mentioned at the beginning of this chapter exemplify this trend. It is interesting to note that the silicon chip area of early single-transceiver systems was

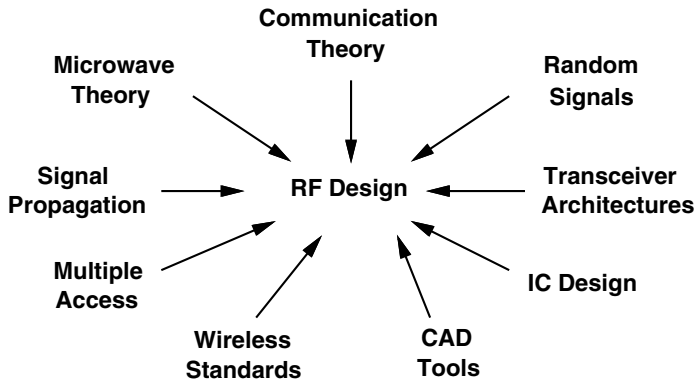


Figure 1.2 *Various disciplines necessary in RF design.*

1. The transit frequency is defined as the frequency at which the small-signal current gain of a device falls to unity.

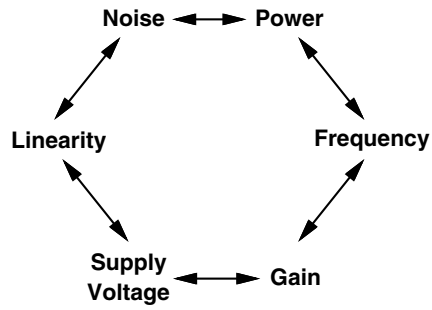


Figure 1.3 *RF design hexagon.*

dominated by the digital baseband processor, allowing RF and analog designers some latitude in the choice of their circuit and device topologies. In today's designs, however, the multiple transceivers tend to occupy a *larger* area than the baseband processor, requiring that RF and analog sections be designed with much care about their area consumption. For example, while on-chip spiral inductors (which have a large footprint) were utilized in abundance in older systems, they are now used only sparingly.

1.3 THE BIG PICTURE

The objective of an RF transceiver is to transmit and receive information. We envision that the transmitter (TX) somehow processes the voice or data signal and applies the result to the antenna [Fig. 1.4(a)]. Similarly, the receiver (RX) senses the signal picked up by the antenna and processes it so as to reconstruct the original voice or data information. Each black box in Fig. 1.4(a) contains a great many functions, but we can readily make two observations: (1) the TX must drive the antenna with a high power level so that the transmitted signal is strong enough to reach far distances, and (2) the RX may sense a small signal (e.g., when a cell phone is used in the basement of a building) and must first amplify the signal with low noise. We now architect our transceiver as shown in Fig. 1.4(b), where the signal to be transmitted is first applied to a "modulator" or "upconverter" so that its center frequency goes from zero to, say, $f_c = 2.4$ GHz. The result drives the antenna through a "power amplifier" (PA). On the receiver side, the signal is sensed by a "low-noise amplifier" (LNA) and subsequently by a "downconverter" or "demodulator" (also known as a "detector").

The upconversion and downconversion paths in Fig. 1.4(b) are driven by an oscillator, which itself is controlled by a "frequency synthesizer." Figure 1.4(c) shows the overall transceiver.[2] The system looks deceptively simple, but we will need the next 900 pages to cover its RF sections. And perhaps another 900 pages to cover the analog-to-digital and digital-to-analog converters.

2. In some cases, the modulator and the upconverter are one and the same. In some other cases, the modulation is performed in the digital domain before upconversion. Most receivers demodulate and detect the signal digitally, requiring only a downconverter in the analog domain.

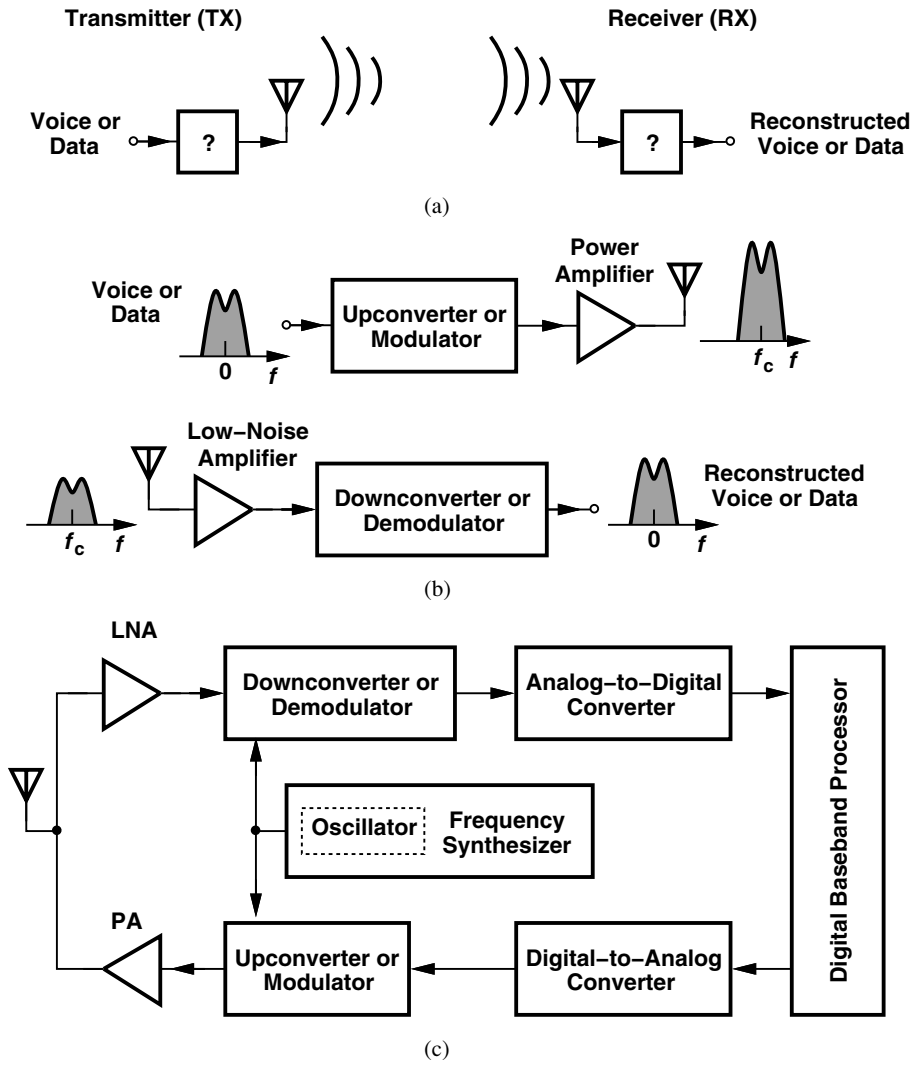


Figure 1.4 *(a) Simple view of RF communication, (b) more complete view, (c) generic RF transceiver.*

REFERENCES

[1] T. Yamawaki et al., "A 2.7-V GSM RF Transceiver IC," *IEEE J. Solid-State Circuits,* vol. 32, pp. 2089–2096, Dec. 1997.

[2] D. Kaczman et al., "A Single-Chip 10-Band WCDMA/HSDPA 4-Band GSM/EDGE SAW-less CMOS Receiver with DigRF 3G Interface and +90-dBm IIP2," *IEEE J. Solid-State Circuits,* vol. 44, pp. 718–739, March 2009.

[3] M. Banu, "MOS Oscillators with Multi-Decade Tuning Range and Gigahertz Maximum Speed," *IEEE J. Solid-State Circuits,* vol. 23, pp. 474–479, April 1988.

[4] B. Razavi et al., "A 3-GHz 25-mW CMOS Phase-Locked Loop," *Dig. of Symposium on VLSI Circuits*, pp. 131–132, June 1994.

[5] M. Soyuer et al., "A 3-V 4-GHz nMOS Voltage-Controlled Oscillator with Integrated Resonator," *IEEE J. Solid-State Circuits,* vol. 31, pp. 2042–2045, Dec. 1996.

[6] B. Kleveland et al., "Monolithic CMOS Distributed Amplifier and Oscillator," *ISSCC Dig. Tech. Papers,* pp. 70–71, Feb. 1999.

[7] H. Wang, "A 50-GHz VCO in 0.25-μm CMOS," *ISSCC Dig. Tech. Papers,* pp. 372–373, Feb. 2001.

[8] L. Franca-Neto, R. Bishop, and B. Bloechel, "64 GHz and 100 GHz VCOs in 90 nm CMOS Using Optimum Pumping Method," ISSCC Dig. Tech. Papers, pp. 444–445, Feb. 2004.

[9] E. Seok et al., "A 410GHz CMOS Push-Push Oscillator with an On-Chip Patch Antenna" *ISSCC Dig. Tech. Papers,* pp. 472–473, Feb. 2008.

[10] B. Razavi, "A 300-GHz Fundamental Oscillator in 65-nm CMOS Technology," *Symposium on VLSI Circuits Dig. Of Tech. Papers,* pp. 113–114, June 2010.

CHAPTER

2

BASIC CONCEPTS IN RF DESIGN

RF design draws upon many concepts from a variety of fields, including signals and systems, electromagnetics and microwave theory, and communications. Nonetheless, RF design has developed its own analytical methods and its own language. For example, while the nonlinear behavior of analog circuits may be characterized by "harmonic distortion," that of RF circuits is quantified by very different measures.

This chapter deals with general concepts that prove essential to the analysis and design of RF circuits, closing the gaps with respect to other fields such as analog design, microwave theory, and communication systems. The outline is shown below.

Nonlinearity
- Harmonic Distortion
- Compression
- Intermodulation
- Dynamic Nonlinear Systems

Noise
- Noise Spectrum
- Device Noise
- Noise in Circuits

Impedance Transformation
- Series–Parallel Conversion
- Matching Networks
- S–Parameters

2.1 GENERAL CONSIDERATIONS

2.1.1 Units in RF Design

RF design has traditionally employed certain units to express gains and signal levels. It is helpful to review these units at the outset so that we can comfortably use them in our subsequent studies.

The voltage gain, V_{out}/V_{in}, and power gain, P_{out}/P_{in}, are expressed in decibels (dB):

$$A_V|_{\text{dB}} = 20\log\frac{V_{out}}{V_{in}} \tag{2.1}$$

$$A_P|_{\text{dB}} = 10\log\frac{P_{out}}{P_{in}}. \tag{2.2}$$

These two quantities are equal (in dB) only if the input and output voltages appear across *equal* impedances. For example, an amplifier having an input resistance of R_0 (e.g., 50 Ω) and driving a load resistance of R_0 satisfies the following equation:

$$A_P|_{\text{dB}} = 10\log\frac{\frac{V_{out}^2}{R_0}}{\frac{V_{in}^2}{R_0}} \tag{2.3}$$

$$= 20\log\frac{V_{out}}{V_{in}} \tag{2.4}$$

$$= A_V|_{\text{dB}}, \tag{2.5}$$

where V_{out} and V_{in} are rms values. In many RF systems, however, this relationship does not hold because the input and output impedances are not equal.

The absolute signal levels are often expressed in dBm rather than in watts or volts. Used for power quantities, the unit dBm refers to "dB's above 1 mW." To express the signal power, P_{sig}, in dBm, we write

$$P_{sig}|_{\text{dBm}} = 10\log\left(\frac{P_{sig}}{1\text{ mW}}\right). \tag{2.6}$$

Example 2.1

An amplifier senses a sinusoidal signal and delivers a power of 0 dBm to a load resistance of 50 Ω. Determine the peak-to-peak voltage swing across the load.

Solution:

Since 0 dBm is equivalent to 1 mW, for a sinusoidal having a peak-to-peak amplitude of V_{pp} and hence an rms value of $V_{pp}/(2\sqrt{2})$, we write

$$\frac{V_{pp}^2}{8R_L} = 1\text{ mW}, \tag{2.7}$$

where $R_L = 50\ \Omega$. Thus,

$$V_{pp} = 632\text{ mV}. \tag{2.8}$$

This is an extremely useful result, as demonstrated in the next example.

Example 2.2

A GSM receiver senses a narrowband (modulated) signal having a level of −100 dBm. If the front-end amplifier provides a voltage gain of 15 dB, calculate the peak-to-peak voltage swing at the output of the amplifier.

Example 2.2 (*Continued*)

Solution:

Since the amplifier output *voltage* swing is of interest, we first convert the received signal level to voltage. From the previous example, we note that $-100\,\text{dBm}$ is $100\,\text{dB}$ below $632\,\text{mV}_{\text{pp}}$. Also, $100\,\text{dB}$ for voltage quantities is equivalent to 10^5. Thus, $-100\,\text{dBm}$ is equivalent to $6.32\ \mu\text{V}_{\text{pp}}$. This input level is amplified by $15\,\text{dB}$ (≈ 5.62), resulting in an output swing of $35.5\ \mu\text{V}_{\text{pp}}$.

The reader may wonder why the output *voltage* of the amplifier is of interest in the above example. This may occur if the circuit following the amplifier does not present a 50-Ω input impedance, and hence the power gain and voltage gain are not equal in dB. In fact, the next stage may exhibit a purely *capacitive* input impedance, thereby requiring no signal "power." This situation is more familiar in analog circuits wherein one stage drives the gate of the transistor in the next stage. As explained in Chapter 5, in most integrated RF systems, we prefer voltage quantities to power quantities so as to avoid confusion if the input and output impedances of cascade stages are unequal or contain negligible real parts.

The reader may also wonder why we were able to assume $0\,\text{dBm}$ is equivalent to $632\,\text{mV}_{\text{pp}}$ in the above example even though the signal is not a pure sinusoid. After all, only for a sinusoid can we assume that the rms value is equal to the peak-to-peak value divided by $2\sqrt{2}$. Fortunately, for a narrowband 0-dBm signal, it is still possible to approximate the (average) peak-to-peak swing as 632 mV.

Although dBm is a unit of power, we sometimes use it at interfaces that do not necessarily entail power transfer. For example, consider the case shown in Fig. 2.1(a), where the LNA drives a purely-capacitive load with a $632\text{-mV}_{\text{pp}}$ swing, delivering no average power. We mentally attach an ideal voltage buffer to node X and drive a 50-Ω load [Fig. 2.1(b)]. We then say that the signal at node X has a level of 0 dBm, tacitly meaning that *if* this signal were applied to a 50-Ω load, *then* it would deliver 1 mW.

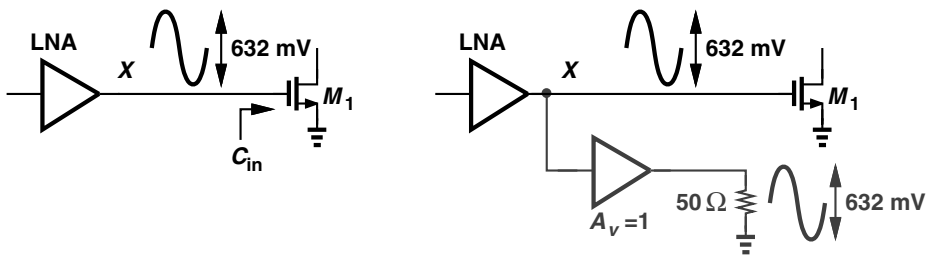


Figure 2.1 *(a) LNA driving a capacitive impedance, (b) use of fictitious buffer to visualize the signal level in dBm.*

2.1.2 Time Variance

A system is linear if its output can be expressed as a linear combination (superposition) of responses to individual inputs. More specifically, if the outputs in response to inputs $x_1(t)$

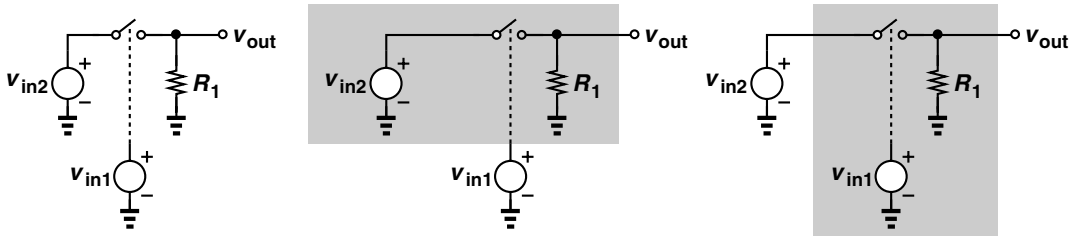


Figure 2.2 *(a) Simple switching circuit, (b) system with V_{in1} as the input, (c) system with V_{in2} as the input.*

and $x_2(t)$ can be respectively expressed as

$$y_1(t) = f[x_1(t)] \tag{2.9}$$

$$y_2(t) = f[x_2(t)], \tag{2.10}$$

then,

$$ay_1(t) + by_2(t) = f[ax_1(t) + bx_2(t)], \tag{2.11}$$

for arbitrary values of a and b. Any system that does not satisfy this condition is nonlinear. Note that, according to this definition, nonzero initial conditions or dc offsets also make a system nonlinear, but we often relax the rule to accommodate these two effects.

Another attribute of systems that may be confused with nonlinearity is time variance. A system is time-invariant if a time shift in its input results in the same time shift in its output. That is, if $y(t) = f[x(t)]$, then $y(t - \tau) = f[x(t - \tau)]$ for arbitrary τ.

As an example of an RF circuit in which time variance plays a critical role and must not be confused with nonlinearity, let us consider the simple switching circuit shown in Fig. 2.2(a). The control terminal of the switch is driven by $v_{in1}(t) = A_1 \cos \omega_1 t$ and the input terminal by $v_{in2}(t) = A_2 \cos \omega_2 t$. We assume the switch is on if $v_{in1} > 0$ and off otherwise. Is this system nonlinear or time-variant? If, as depicted in Fig. 2.2(b), the input of interest is v_{in1} (while v_{in2} is part of the system and still equal to $A_2 \cos \omega_2 t$), then the system is nonlinear because the control is only sensitive to the polarity of v_{in1} and independent of its amplitude. This system is also time-variant because the output depends on v_{in2}. For example, if v_{in1} is constant and positive, then $v_{out}(t) = v_{in2}(t)$, and if v_{in1} is constant and negative, then $v_{out}(t) = 0$ (why?).

Now consider the case shown in Fig. 2.2(c), where the input of interest is v_{in2} (while v_{in1} remains part of the system and still equal to $A_1 \cos \omega_1 t$). This system is linear with respect to v_{in2}. For example, doubling the amplitude of v_{in2} directly doubles that of v_{out}. The system is also time-variant due to the effect of v_{in1}.

Example 2.3

Plot the output waveform of the circuit in Fig. 2.2(a) if $v_{in1} = A_1 \cos \omega_1 t$ and $v_{in2} = A_2 \cos(1.25\omega_1 t)$.

Example 2.3 (*Continued*)

Solution:

As shown in Fig. 2.3, v_{out} tracks v_{in2} if $v_{in1} > 0$ and is pulled down to zero by R_1 if $v_{in1} < 0$. That is, v_{out} is equal to the product of v_{in2} and a square wave toggling between 0 and 1.

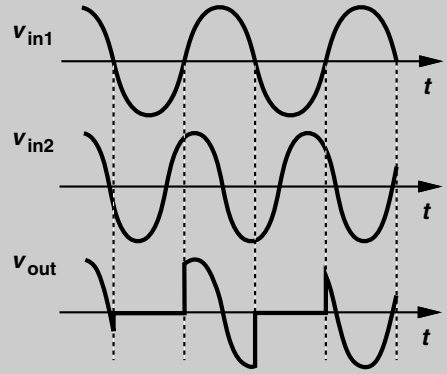


Figure 2.3 *Input and output waveforms.*

The circuit of Fig. 2.2(a) is an example of RF "mixers." We will study such circuits in Chapter 6 extensively, but it is important to draw several conclusions from the above study. First, statements such as "switches are nonlinear" are ambiguous. Second, a linear system *can* generate frequency components that do not exist in the input signal—the system only need be time-variant. From Example 2.3,

$$v_{out}(t) = v_{in2}(t) \cdot S(t), \tag{2.12}$$

where $S(t)$ denotes a square wave toggling between 0 and 1 with a frequency of $f_1 = \omega_1/(2\pi)$. The output spectrum is therefore given by the convolution of the spectra of $v_{in2}(t)$ and $S(t)$. Since the spectrum of a square wave is equal to a train of impulses whose amplitudes follow a sinc envelope, we have

$$V_{out}(f) = V_{in2}(f) * \sum_{n=-\infty}^{+\infty} \frac{\sin(n\pi/2)}{n\pi} \delta\left(f - \frac{n}{T_1}\right) \tag{2.13}$$

$$= \sum_{n=-\infty}^{+\infty} \frac{\sin(n\pi/2)}{n\pi} V_{in2}\left(f - \frac{n}{T_1}\right), \tag{2.14}$$

where $T_1 = 2\pi/\omega_1$. This operation is illustrated in Fig. 2.4 for a V_{in2} spectrum located around zero frequency.[1]

1. It is helpful to remember that, for $n = 1$, each impulse in the above summation has an area of $1/\pi$ and the corresponding sinusoid, a *peak amplitude* of $2/\pi$.

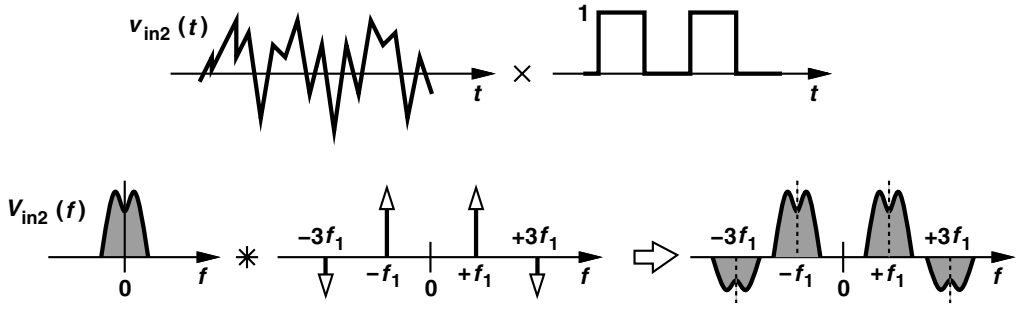


Figure 2.4 *Multiplication in the time domain and corresponding convolution in the frequency domain.*

2.1.3 Nonlinearity

A system is called "memoryless" or "static" if its output does not depend on the past values of its input (or the past values of the output itself). For a memoryless linear system, the input/output characteristic is given by

$$y(t) = \alpha x(t), \tag{2.15}$$

where α is a function of time if the system is time-variant [e.g., Fig. 2.2(c)]. For a memoryless nonlinear system, the input/output characteristic can be approximated with a polynomial,

$$y(t) = \alpha_0 + \alpha_1 x(t) + \alpha_2 x^2(t) + \alpha_3 x^3(t) + \cdots, \tag{2.16}$$

where α_j may be functions of time if the system is time-variant. Figure 2.5 shows a common-source stage as an example of a memoryless nonlinear circuit (at low frequencies). If M_1 operates in the saturation region and can be approximated as a square-law device, then

$$V_{out} = V_{DD} - I_D R_D \tag{2.17}$$

$$= V_{DD} - \frac{1}{2}\mu_n C_{ox}\frac{W}{L}(V_{in} - V_{TH})^2 R_D. \tag{2.18}$$

In this idealized case, the circuit displays only second-order nonlinearity.

The system described by Eq. (2.16) has "odd symmetry" if $y(t)$ is an odd function of $x(t)$, i.e., if the response to $-x(t)$ is the negative of that to $+x(t)$. This occurs if $\alpha_j = 0$ for even j. Such a system is sometimes called "balanced," as exemplified by the differential

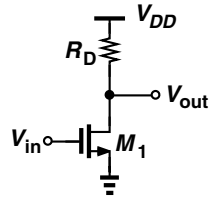


Figure 2.5 *Common-source stage.*

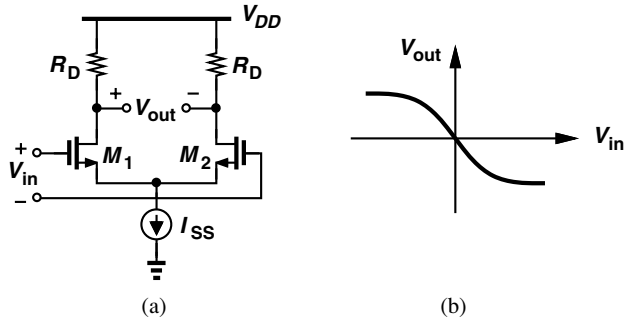


Figure 2.6 *(a) Differential pair and (b) its input/output characteristic.*

pair shown in Fig. 2.6(a). Recall from basic analog design that by virtue of symmetry, the circuit exhibits the characteristic depicted in Fig. 2.6(b) if the differential input varies from very negative values to very positive values.

Example 2.4

For square-law MOS transistors operating in saturation, the characteristic of Fig. 2.6(b) can be expressed as [1]

$$V_{out} = -\frac{1}{2}\mu_n C_{ox}\frac{W}{L}V_{in}\sqrt{\frac{4I_{SS}}{\mu_n C_{ox}\frac{W}{L}} - V_{in}^2}R_D. \tag{2.19}$$

If the differential input is small, approximate the characteristic by a polynomial.

Solution:

Factoring $4I_{SS}/(\mu_n C_{ox}W/L)$ out of the square root and assuming

$$V_{in}^2 \ll \frac{4I_{SS}}{\mu_n C_{ox}\frac{W}{L}}, \tag{2.20}$$

we use the approximation $\sqrt{1-\epsilon} \approx 1 - \epsilon/2$ to write

$$V_{out} \approx -\sqrt{\mu_n C_{ox}\frac{W}{L}I_{SS}}V_{in}\left(1 - \frac{\mu_n C_{ox}\frac{W}{L}}{8I_{SS}}V_{in}^2\right)R_D \tag{2.21}$$

$$\approx -\sqrt{\mu_n C_{ox}\frac{W}{L}I_{SS}}R_D V_{in} + \frac{\left(\mu_n C_{ox}\frac{W}{L}\right)^{3/2}}{8\sqrt{I_{SS}}}R_D V_{in}^3. \tag{2.22}$$

The first term on the right-hand side represents linear operation, revealing the small-signal voltage gain of the circuit ($-g_m R_D$). Due to symmetry, even-order nonlinear terms are absent. Interestingly, square-law devices yield a *third-order* characteristic in this case. We return to this point in Chapter 5.

A system is called "dynamic" if its output depends on the past values of its input(s) or output(s). For a linear, time-invariant, dynamic system,

$$y(t) = h(t) * x(t), \tag{2.23}$$

where $h(t)$ denotes the impulse response. If a dynamic system is linear but time-variant, its impulse response depends on the time origin; if $\delta(t)$ yields $h(t)$, then $\delta(t - \tau)$ produces $h(t, \tau)$. Thus,

$$y(t) = h(t, \tau) * x(t). \tag{2.24}$$

Finally, if a system is both nonlinear and dynamic, then its impulse response can be approximated by a Volterra series. This is described in Section 2.8.

2.2 EFFECTS OF NONLINEARITY

While analog and RF circuits can be approximated by a linear model for small-signal operation, nonlinearities often lead to interesting and important phenomena that are not predicted by small-signal models. In this section, we study these phenomena for memoryless systems whose input/output characteristic can be approximated by[2]

$$y(t) \approx \alpha_1 x(t) + \alpha_2 x^2(t) + \alpha_3 x^3(t). \tag{2.25}$$

The reader is cautioned, however, that the effect of storage elements (dynamic nonlinearity) and higher-order nonlinear terms must be carefully examined to ensure (2.25) is a plausible representation. Section 2.7 deals with the case of dynamic nonlinearity. We may consider α_1 as the small-signal gain of the system because the other two terms are negligible for small input swings. For example, $\alpha_1 = -\sqrt{\mu_n C_{ox}(W/L) I_{SS}} R_D$ in Eq. (2.22).

The nonlinearity effects described in this section primarily arise from the third-order term in Eq. (2.25). The second-order term too manifests itself in certain types of receivers and is studied in Chapter 4.

2.2.1 Harmonic Distortion

If a sinusoid is applied to a nonlinear system, the output generally exhibits frequency components that are integer multiples ("harmonics") of the input frequency. In Eq. (2.25), if $x(t) = A \cos \omega t$, then

$$y(t) = \alpha_1 A \cos \omega t + \alpha_2 A^2 \cos^2 \omega t + \alpha_3 A^3 \cos^3 \omega t \tag{2.26}$$

$$= \alpha_1 A \cos \omega t + \frac{\alpha_2 A^2}{2}(1 + \cos 2\omega t) + \frac{\alpha_3 A^3}{4}(3 \cos \omega t + \cos 3\omega t) \tag{2.27}$$

$$= \frac{\alpha_2 A^2}{2} + \left(\alpha_1 A + \frac{3\alpha_3 A^3}{4}\right) \cos \omega t + \frac{\alpha_2 A^2}{2} \cos 2\omega t + \frac{\alpha_3 A^3}{4} \cos 3\omega t. \tag{2.28}$$

2. Note that this expression should be considered as a fit across the signal swings of interest rather than as a Taylor expansion in the vicinity of $x = 0$. These two views may yield slightly different values for α_j.

In Eq. (2.28), the first term on the right-hand side is a dc quantity arising from second-order nonlinearity, the second is called the "fundamental," the third is the second harmonic, and the fourth is the third harmonic. We sometimes say that even-order nonlinearity introduces dc offsets.

From the above expansion, we make two observations. First, even-order harmonics result from α_j with even j, and vanish if the system has odd symmetry, i.e., if it is fully differential. In reality, however, random mismatches corrupt the symmetry, yielding finite even-order harmonics. Second, in (2.28) the amplitudes of the second and third harmonics are proportional to A^2 and A^3, respectively, i.e., we say the nth harmonic grows in proportion to A^n.

In many RF circuits, harmonic distortion is unimportant or an irrelevant indicator of the effect of nonlinearity. For example, an amplifier operating at 2.4 GHz produces a second harmonic at 4.8 GHz, which is greatly suppressed if the circuit has a narrow bandwidth. Nonetheless, harmonics must always be considered carefully before they are dismissed. The following examples illustrate this point.

Example 2.5

An analog multiplier "mixes" its two inputs as shown in Fig. 2.7, ideally producing $y(t) = kx_1(t)x_2(t)$, where k is a constant.[3] Assume $x_1(t) = A_1 \cos \omega_1 t$ and $x_2(t) = A_2 \cos \omega_2 t$.

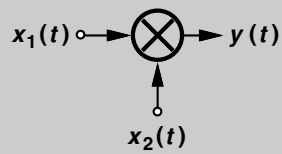


Figure 2.7 *Analog multiplier.*

(a) If the mixer is ideal, determine the output frequency components.
(b) If the input port sensing $x_2(t)$ suffers from third-order nonlinearity, determine the output frequency components.

Solution:

(a) We have

$$y(t) = k(A_1 \cos \omega_1 t)(A_2 \cos \omega_2 t) \tag{2.29}$$

$$= \frac{kA_1A_2}{2} \cos(\omega_1 + \omega_2)t + \frac{kA_1A_2}{2} \cos(\omega_1 - \omega_2)t. \tag{2.30}$$

The output thus contains the sum and difference frequencies. These may be considered "desired" components.

(*Continues*)

3. The factor k is necessary to ensure a proper dimension for $y(t)$.

Example 2.5 (*Continued*)

(b) Representing the third harmonic of $x_2(t)$ by $(\alpha_3 A_2^3/4)\cos 3\omega_2 t$, we write

$$y(t) = k(A_1 \cos\omega_1 t)\left(A_2 \cos\omega_2 t + \frac{\alpha_3 A_2^3}{4}\cos 3\omega_2 t\right) \tag{2.31}$$

$$= \frac{kA_1A_2}{2}\cos(\omega_1 + \omega_2)t + \frac{kA_1A_2}{2}\cos(\omega_1 - \omega_2)t + \frac{k\alpha_3 A_1 A_2^3}{8}\cos(\omega_1 + 3\omega_2)t + \frac{k\alpha_3 A_1 A_2^3}{8}\cos(\omega_1 - 3\omega_2)t. \tag{2.32}$$

The mixer now produces two "spurious" components at $\omega_1 + 3\omega_2$ and $\omega_1 - 3\omega_2$, one or both of which often prove problematic. For example, if $\omega_1 = 2\pi \times (850\text{ MHz})$ and $\omega_2 = 2\pi \times (900\text{ MHz})$, then $|\omega_1 - 3\omega_2| = 2\pi \times (1850\text{ MHz})$, an "undesired" component that is difficult to filter because it lies close to the desired component at $\omega_1 + \omega_2 = 2\pi \times (1750\text{ MHz})$.

Example 2.6

The transmitter in a 900-MHz GSM cellphone delivers 1 W of power to the antenna. Explain the effect of the harmonics of this signal.

Solution:

The second harmonic falls within another GSM cell phone band around 1800 MHz and must be sufficiently small to negligibly impact the other users in that band. The third, fourth, and fifth harmonics do not coincide with any popular bands but must still remain below a certain level imposed by regulatory organizations in each country. The sixth harmonic falls in the 5-GHz band used in wireless local area networks (WLANs), e.g., in laptops. Figure 2.8 summarizes these results.

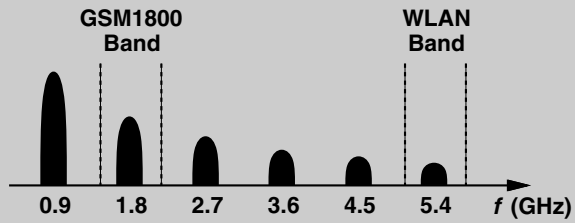


Figure 2.8 *Summary of harmonic components.*

2.2.2 Gain Compression

The small-signal gain of circuits is usually obtained with the assumption that harmonics are negligible. However, our formulation of harmonics, as expressed by Eq. (2.28), indicates

that the gain experienced by $A\cos\omega t$ is equal to $\alpha_1 + 3\alpha_3 A^2/4$ and hence varies appreciably as A becomes larger.[4] We must then ask, do α_1 and α_3 have the same sign or opposite signs? Returning to the third-order polynomial in Eq. (2.25), we note that if $\alpha_1\alpha_3 > 0$, then $\alpha_1 x + \alpha_3 x^3$ overwhelms $\alpha_2 x^2$ for large x regardless of the sign of α_2, yielding an "expansive" characteristic [Fig. 2.9(a)]. For example, an ideal bipolar transistor operating in the forward active region produces a collector current in proportion to $\exp(V_{BE}/V_T)$, exhibiting expansive behavior. On the other hand, if $\alpha_1\alpha_3 < 0$, the term $\alpha_3 x^3$ "bends" the characteristic for sufficiently large x [Fig. 2.9(b)], leading to "compressive" behavior, i.e., a decreasing gain as the input amplitude increases. For example, the differential pair of Fig. 2.6(a) suffers from compression as the second term in (2.22) becomes comparable with the first. Since most RF circuits of interest are compressive, we hereafter focus on this type.

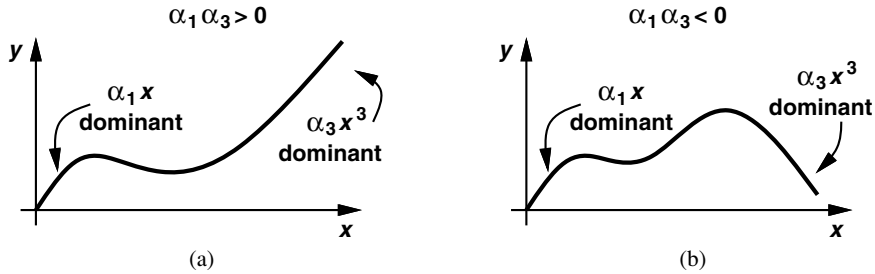


Figure 2.9 *(a) Expansive and (b) compressive characteristics.*

With $\alpha_1\alpha_3 < 0$, the gain experienced by $A\cos\omega t$ in Eq. (2.28) falls as A rises. We quantify this effect by the "1-dB compression point," defined as the input signal level that causes the gain to drop by 1 dB. If plotted on a log-log scale as a function of the input level, the output level, A_{out}, falls below its ideal value by 1 dB at the 1-dB compression point, $A_{in,1dB}$ (Fig. 2.10). Note that (a) A_{in} and A_{out} are voltage quantities here, but compression can also be expressed in terms of power quantities; (b) 1-dB compression may also be specified in terms of the output level at which it occurs, $A_{out,1dB}$. The input and output compression points typically prove relevant in the receive path and the transmit path, respectively.

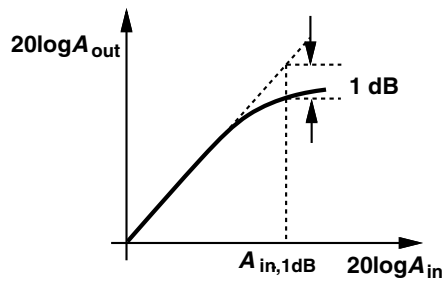


Figure 2.10 *Definition of 1-dB compression point.*

4. This effect is akin to the fact that nonlinearity can also be viewed as variation of the *slope* of the input/output characteristic with the input level.

To calculate the input 1-dB compression point, we equate the compressed gain, $\alpha_1 + (3\alpha_3/4)A^2_{in,1dB}$, to 1 dB less than the ideal gain, α_1:

$$20\log\left|\alpha_1 + \frac{3}{4}\alpha_3 A^2_{in,1dB}\right| = 20\log|\alpha_1| - 1\ \text{dB}. \tag{2.33}$$

It follows that

$$A_{in,1dB} = \sqrt{0.145\left|\frac{\alpha_1}{\alpha_3}\right|}. \tag{2.34}$$

Note that Eq. (2.34) gives the *peak* value (rather than the peak-to-peak value) of the input. Also denoted by P_{1dB}, the 1-dB compression point is typically in the range of -20 to -25 dBm (63.2 to 35.6 mV_{pp} in 50-Ω system) at the input of RF receivers. We use the notations A_{1dB} and P_{1dB} interchangeably in this book. Whether they refer to the input or the output will be clear from the context or specified explicitly. While gain compression by 1 dB seems arbitrary, the 1-dB compression point represents a 10% reduction in the gain and is widely used to characterize RF circuits and systems.

Why does compression matter? After all, it appears that if a signal is so large as to reduce the gain of a receiver, then it must lie well above the receiver noise and be easily detectable. In fact, for some modulation schemes, this statement holds and compression of the receiver would seem benign. For example, as illustrated in Fig. 2.11(a), a frequency-modulated signal carries no information in its amplitude and hence tolerates compression (i.e., amplitude limiting). On the other hand, modulation schemes that contain information in the amplitude are distorted by compression [Fig. 2.11(b)]. This issue manifests itself in both receivers and transmitters.

Another adverse effect arising from compression occurs if a large *interferer* accompanies the received signal [Fig. 2.12(a)]. In the time domain, the small desired signal is superimposed on the large interferer. Consequently, the receiver gain is reduced by the large excursions produced by the interferer even though the desired signal itself is small

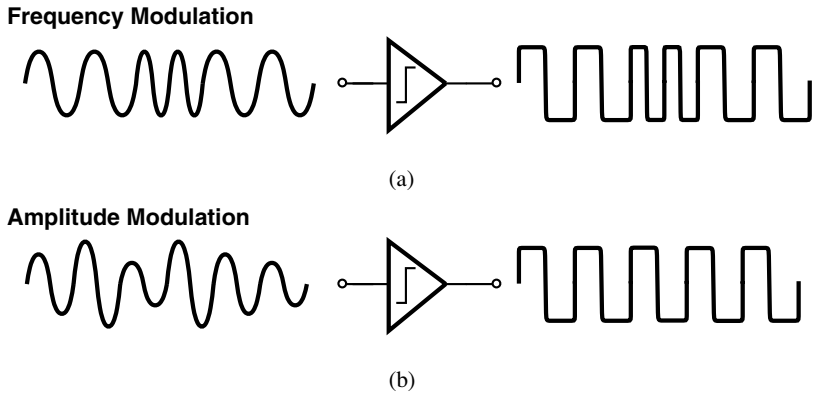


Figure 2.11 *Effect of compressive nonlinearity on (a) FM and (b) AM waveforms.*

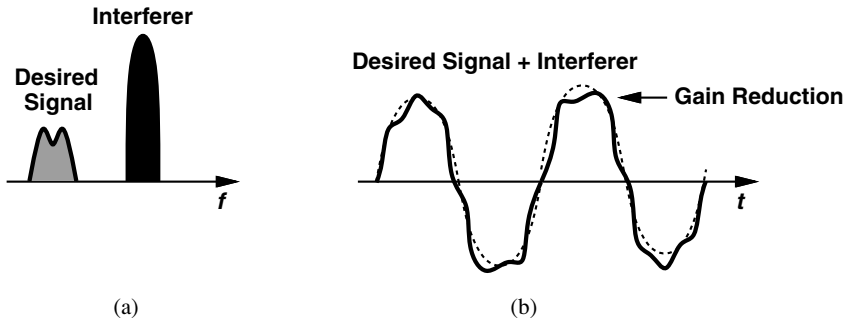


Figure 2.12 *(a) Interferer accompanying signal, (b) effect in time domain.*

[Fig. 2.12(b)]. Called "desensitization," this phenomenon lowers the signal-to-noise ratio (SNR) at the receiver output and proves critical even if the signal contains no amplitude information.

To quantify desensitization, let us assume $x(t) = A_1 \cos \omega_1 t + A_2 \cos \omega_2 t$, where the first and second terms represent the desired component and the interferer, respectively. With the third-order characteristic of Eq. (2.25), the output appears as

$$y(t) = \left(\alpha_1 + \frac{3}{4}\alpha_3 A_1^2 + \frac{3}{2}\alpha_3 A_2^2\right) A_1 \cos \omega_1 t + \cdots . \tag{2.35}$$

Note that α_2 is absent in compression. For $A_1 \ll A_2$, this reduces to

$$y(t) = \left(\alpha_1 + \frac{3}{2}\alpha_3 A_2^2\right) A_1 \cos \omega_1 t + \cdots . \tag{2.36}$$

Thus, the gain experienced by the desired signal is equal to $\alpha_1 + 3\alpha_3 A_2^2/2$, a decreasing function of A_2 if $\alpha_1 \alpha_3 < 0$. In fact, for sufficiently large A_2, the gain drops to zero, and we say the signal is "blocked." In RF design, the term "blocking signal" or "blocker" refers to interferers that desensitize a circuit even if they do not reduce the gain to zero. Some RF receivers must be able to withstand blockers that are 60 to 70 dB greater than the desired signal.

Example 2.7

A 900-MHz GSM transmitter delivers a power of 1 W to the antenna. By how much must the second harmonic of the signal be suppressed (filtered) so that it does not desensitize a 1.8-GHz receiver having $P_{1dB} = -25$ dBm? Assume the receiver is 1 m away (Fig. 2.13) and the 1.8-GHz signal is attenuated by 10 dB as it propagates across this distance.

(*Continues*)

Example 2.7 (*Continued*)

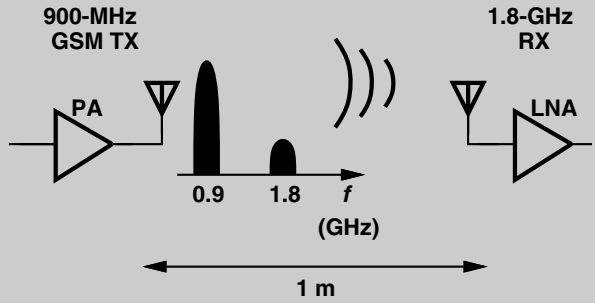


Figure 2.13 *TX and RX in a cellular system.*

Solution:

The output power at 900 MHz is equal to +30 dBm. With an attenuation of 10 dB, the second harmonic must not exceed −15 dBm at the transmitter antenna so that it is below P_{1dB} of the receiver. Thus, the second harmonic must remain at least 45 dB below the fundamental at the TX output. In practice, this interference must be another several dB lower to ensure the RX does not compress.

2.2.3 Cross Modulation

Another phenomenon that occurs when a weak signal and a strong interferer pass through a nonlinear system is the *transfer* of modulation from the interferer to the signal. Called "cross modulation," this effect is exemplified by Eq. (2.36), where variations in A_2 affect the amplitude of the signal at ω_1. For example, suppose that the interferer is an amplitude-modulated signal, $A_2(1 + m\cos\omega_m t)\cos\omega_2 t$, where m is a constant and ω_m denotes the modulating frequency. Equation (2.36) thus assumes the following form:

$$y(t) = \left[\alpha_1 + \frac{3}{2}\alpha_3 A_2^2\left(1 + \frac{m^2}{2} + \frac{m^2}{2}\cos 2\omega_m t + 2m\cos\omega_m t\right)\right] A_1\cos\omega_1 t + \cdots . \quad (2.37)$$

In other words, the desired signal at the output suffers from amplitude modulation at ω_m and $2\omega_m$. Figure 2.14 illustrates this effect.

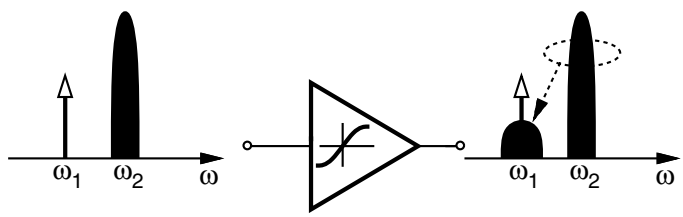


Figure 2.14 *Cross modulation.*

Example 2.8

Suppose an interferer contains phase modulation but not amplitude modulation. Does cross modulation occur in this case?

Solution:

Expressing the input as $x(t) = A_1 \cos \omega_1 t + A_2 \cos(\omega_2 t + \phi)$, where the second term represents the interferer (A_2 is constant but ϕ varies with time), we use the third-order polynomial in Eq. (2.25) to write

$$\begin{aligned} y(t) &= \alpha_1[A_1 \cos \omega_1 t + A_2 \cos(\omega_2 t + \phi)] + \alpha_2[A_1 \cos \omega_1 t + A_2 \cos(\omega_2 t + \phi)]^2 \\ &\quad + \alpha_3[A_1 \cos \omega_1 t + A_2 \cos(\omega_2 t + \phi)]^3. \end{aligned} \tag{2.38}$$

We now note that (1) the second-order term yields components at $\omega_1 \pm \omega_2$ but not at ω_1; (2) the third-order term expansion gives $3\alpha_3 A_1 \cos \omega_1 t A_2^2 \cos^2(\omega_2 t + \phi)$, which, according to $\cos^2 x = (1 + \cos 2x)/2$, results in a component at ω_1. Thus,

$$y(t) = \left(\alpha_1 + \frac{3}{2}\alpha_3 A_2^2\right) A_1 \cos \omega_1 t + \cdots . \tag{2.39}$$

Interestingly, the desired signal at ω_1 does not experience cross modulation. That is, phase-modulated interferers do not cause cross modulation in *memoryless* (static) nonlinear systems. Dynamic nonlinear systems, on the other hand, may not follow this rule.

Cross modulation commonly arises in amplifiers that must simultaneously process many independent signal channels. Examples include cable television transmitters and systems employing "orthogonal frequency division multiplexing" (OFDM). We examine OFDM in Chapter 3.

2.2.4 Intermodulation

Our study of nonlinearity has thus far considered the case of a single signal (for harmonic distortion) or a signal accompanied by one large interferer (for desensitization). Another scenario of interest in RF design occurs if *two* interferers accompany the desired signal. Such a scenario represents realistic situations and reveals nonlinear effects that may not manifest themselves in a harmonic distortion or desensitization test.

If two interferers at ω_1 and ω_2 are applied to a nonlinear system, the output generally exhibits components that are not harmonics of these frequencies. Called "intermodulation" (IM), this phenomenon arises from "mixing" (multiplication) of the two components as their sum is raised to a power greater than unity. To understand how Eq. (2.25) leads to intermodulation, assume $x(t) = A_1 \cos \omega_1 t + A_2 \cos \omega_2 t$. Thus,

$$\begin{aligned} y(t) &= \alpha_1(A_1 \cos \omega_1 t + A_2 \cos \omega_2 t) + \alpha_2(A_1 \cos \omega_1 t + A_2 \cos \omega_2 t)^2 \\ &\quad + \alpha_3(A_1 \cos \omega_1 t + A_2 \cos \omega_2 t)^3. \end{aligned} \tag{2.40}$$

Expanding the right-hand side and discarding the dc terms, harmonics, and components at $\omega_1 \pm \omega_2$, we obtain the following "intermodulation products":

$$\omega = 2\omega_1 \pm \omega_2 : \frac{3\alpha_3 A_1^2 A_2}{4}\cos(2\omega_1 + \omega_2)t + \frac{3\alpha_3 A_1^2 A_2}{4}\cos(2\omega_1 - \omega_2)t \quad (2.41)$$

$$\omega = 2\omega_2 \pm \omega_1 : \frac{3\alpha_3 A_1 A_2^2}{4}\cos(2\omega_2 + \omega_1)t + \frac{3\alpha_3 A_1 A_2^2}{4}\cos(2\omega_2 - \omega_1)t \quad (2.42)$$

and these fundamental components:

$$\omega = \omega_1,\ \omega_2 : \left(\alpha_1 A_1 + \frac{3}{4}\alpha_3 A_1^3 + \frac{3}{2}\alpha_3 A_1 A_2^2\right)\cos\omega_1 t + \left(\alpha_1 A_2 + \frac{3}{4}\alpha_3 A_2^3 + \frac{3}{2}\alpha_3 A_2 A_1^2\right)\cos\omega_2 t \quad (2.43)$$

Figure 2.15 illustrates the results. Among these, the third-order IM products at $2\omega_1 - \omega_2$ and $2\omega_2 - \omega_1$ are of particular interest. This is because, if ω_1 and ω_2 are close to each other, then $2\omega_1 - \omega_2$ and $2\omega_2 - \omega_1$ appear in the vicinity of ω_1 and ω_2. We now explain the significance of this statement.

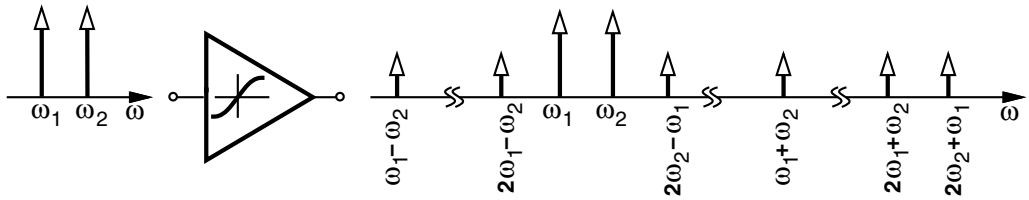


Figure 2.15 *Generation of various intermodulation components in a two-tone test.*

Suppose an antenna receives a small desired signal at ω_0 along with two large interferers at ω_1 and ω_2, providing this combination to a low-noise amplifier (Fig. 2.16). Let us assume that the interferer frequencies happen to satisfy $2\omega_1 - \omega_2 = \omega_0$. Consequently, the intermodulation product at $2\omega_1 - \omega_2$ falls onto the desired channel, corrupting the signal.

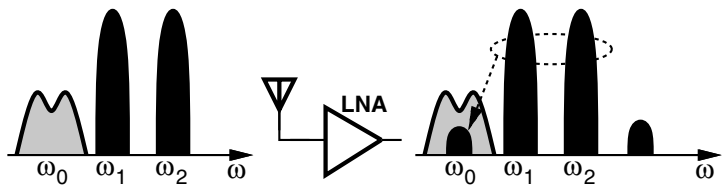


Figure 2.16 *Corruption due to third-order intermodulation.*

Example 2.9

Suppose four Bluetooth users operate in a room as shown in Fig. 2.17. User 4 is in the receive mode and attempts to sense a weak signal transmitted by User 1 at 2.410 GHz.

Example 2.9 (*Continued*)

At the same time, Users 2 and 3 transmit at 2.420 GHz and 2.430 GHz, respectively. Explain what happens.

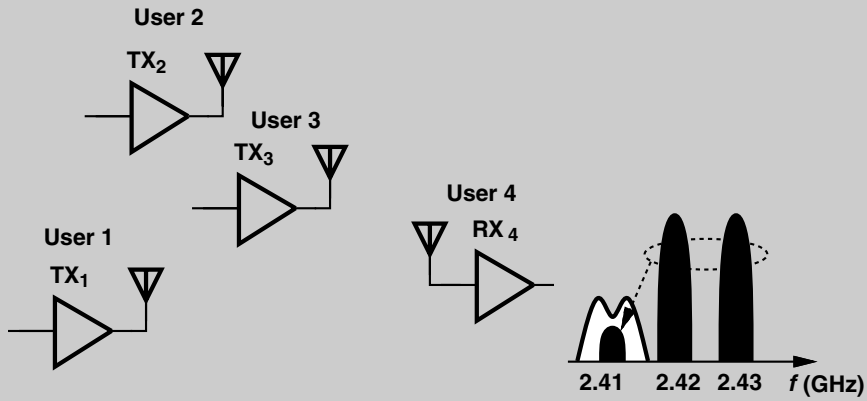


Figure 2.17 *Bluetooth RX in the presence of several transmitters.*

Solution:

Since the frequencies transmitted by Users 1, 2, and 3 happen to be equally spaced, the intermodulation in the LNA of RX_4 corrupts the desired signal at 2.410 GHz.

The reader may raise several questions at this point: (1) In our analysis of intermodulation, we represented the interferers with pure (unmodulated) sinusoids (called "tones") whereas in Figs. 2.16 and 2.17, the interferers are modulated. Are these consistent? (2) Can gain compression and desensitization (P_{1dB}) also model intermodulation, or do we need other measures of nonlinearity? (3) Why can we not simply remove the interferers by filters so that the receiver does not experience intermodulation? We answer the first two here and address the third in Chapter 4.

For narrowband signals, it is sometimes helpful to "condense" their energy into an impulse, i.e., represent them with a tone of equal power [Fig. 2.18(a)]. This approximation must be made judiciously: if applied to study gain compression, it yields reasonably accurate results; on the other hand, if applied to the case of cross modulation, it fails. In intermodulation analyses, we proceed as follows: (a) approximate the interferers with tones, (b) calculate the level of intermodulation products at the output, and (c) mentally convert the intermodulation tones back to modulated components so as to see the corruption.[5] This thought process is illustrated in Fig. 2.18(b).

We now deal with the second question: if the gain is not compressed, then can we say that intermodulation is negligible? The answer is no; the following example illustrates this point.

5. Since a tone contains no randomness, it generally does not corrupt a signal. But a tone appearing in the spectrum of a signal may make the detection difficult.

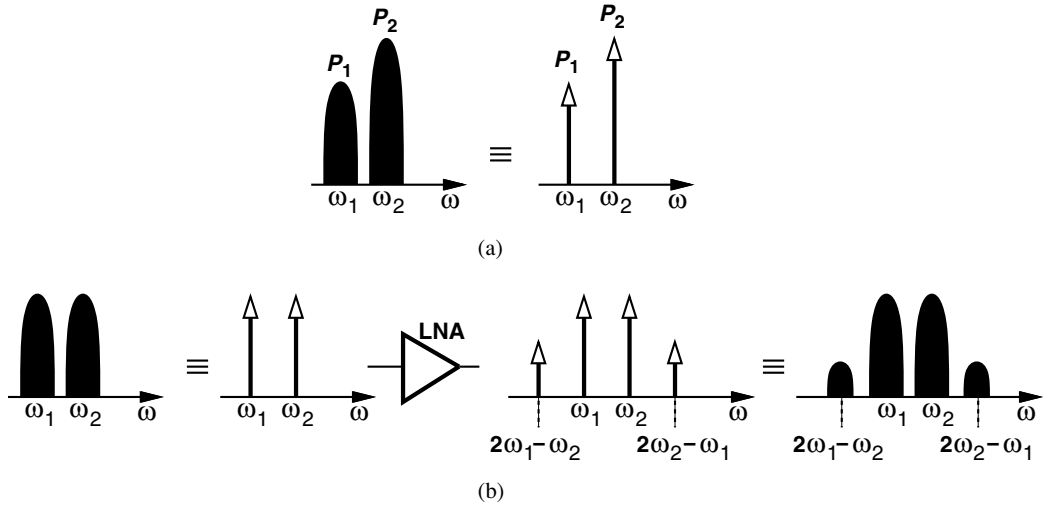


Figure 2.18 *(a) Approximation of modulated signals by impulses, (b) application to intermodulation.*

Example 2.10

A Bluetooth receiver employs a low-noise amplifier having a gain of 10 and an input impedance of 50 Ω. The LNA senses a desired signal level of −80 dBm at 2.410 GHz and two interferers of equal levels at 2.420 GHz and 2.430 GHz. For simplicity, assume the LNA drives a 50-Ω load.

(a) Determine the value of α_3 that yields a P_{1dB} of −30 dBm.

(b) If each interferer is 10 dB below P_{1dB}, determine the corruption experienced by the desired signal at the LNA output.

Solution:

(a) Noting that $-30\,\text{dBm} = 20\,\text{mV}_{\text{pp}} = 10\,\text{mV}_{\text{p}}$, from Eq. (2.34), we have $\sqrt{0.145|\alpha_1/\alpha_3|} = 10\ \text{mV}_{\text{p}}$. Since $\alpha_1 = 10$, we obtain $\alpha_3 = 14,500\,\text{V}^{-2}$.

(b) Each interferer has a level of −40 dBm ($= 6.32\,\text{mV}_{\text{pp}}$). Setting $A_1 = A_2 = 6.32\,\text{mV}_{\text{pp}}/2$ in Eq. (2.41), we determine the amplitude of the IM product at 2.410 GHz as

$$\frac{3\alpha_3 A_1^2 A_2}{4} = 0.343\,\text{mV}_{\text{p}} = -59.3\,\text{dBm}. \tag{2.44}$$

The desired signal is amplified by a factor of $\alpha_1 = 10 = 20$ dB, emerging at the output at a level of −60 dBm. Unfortunately, the IM product is as large as the signal itself even though the LNA does not experience significant compression.

The two-tone test is versatile and powerful because it can be applied to systems with arbitrarily narrow bandwidths. A sufficiently small difference between the two tone frequencies ensures that the IM products also fall within the band, thereby providing a

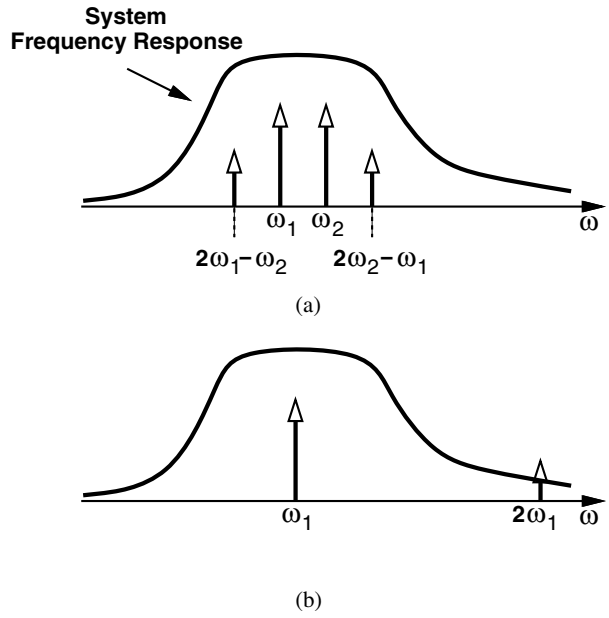


Figure 2.19 *(a) Two-tone and (b) harmonic tests in a narrowband system.*

meaningful view of the nonlinear behavior of the system. Depicted in Fig. 2.19(a), this attribute stands in contrast to harmonic distortion tests, where higher harmonics lie so far away in frequency that they are heavily filtered, making the system appear quite linear [Fig. 2.19(b)].

Third Intercept Point Our thoughts thus far indicate the need for a measure of intermodulation. A common method of IM characterization is the "two-tone" test, whereby two pure sinusoids of *equal* amplitudes are applied to the input. The amplitude of the output IM products is then normalized to that of the fundamentals at the output. Denoting the peak amplitude of each tone by A, we can write the result as

$$\text{Relative IM} = 20\log\left(\frac{3}{4}\frac{\alpha_3}{\alpha_1}A^2\right)\ \text{dBc}, \tag{2.45}$$

where the unit dBc denotes decibels with respect to the "carrier" to emphasize the normalization. Note that, if the amplitude of each input tone increases by 6 dB (a factor of two), the amplitude of the IM products ($\propto A^3$) rises by 18 dB and hence the *relative* IM by 12 dB.[6]

The principal difficulty in specifying the relative IM for a circuit is that it is meaningful only if the value of A is given. From a practical point of view, we prefer a *single* measure that captures the intermodulation behavior of the circuit with no need to know the input level at which the two-tone test is carried out. Fortunately, such a measure exists and is called the "third intercept point" (IP_3).

The concept of IP_3 originates from our earlier observation that, if the amplitude of each tone rises, that of the output IM products increases more sharply ($\propto A^3$). Thus, if we continue to raise A, the amplitude of the IM products eventually becomes *equal* to that

6. It is assumed that no compression occurs so that the output fundamental tones also rise by 6 dB.

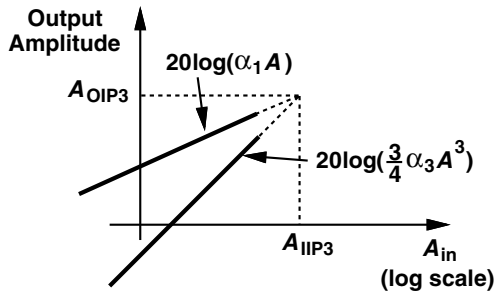


Figure 2.20 *Definition of IP_3 (for voltage quantities).*

of the fundamental tones at the output. As illustrated in Fig. 2.20 on a log-log scale, the input level at which this occurs is called the "input third intercept point" (IIP_3). Similarly, the corresponding output is represented by OIP_3. In subsequent derivations, we denote the input amplitude as A_{IIP3}.

To determine the IIP_3, we simply equate the fundamental and IM amplitudes:

$$|\alpha_1 A_{IIP3}| = \left|\frac{3}{4}\alpha_3 A_{IIP3}^3\right|, \tag{2.46}$$

obtaining

$$A_{IIP3} = \sqrt{\frac{4}{3}\left|\frac{\alpha_1}{\alpha_3}\right|}. \tag{2.47}$$

Interestingly,

$$\frac{A_{IIP3}}{A_{1dB}} = \sqrt{\frac{4}{0.435}} \tag{2.48}$$

$$\approx 9.6\,\text{dB}. \tag{2.49}$$

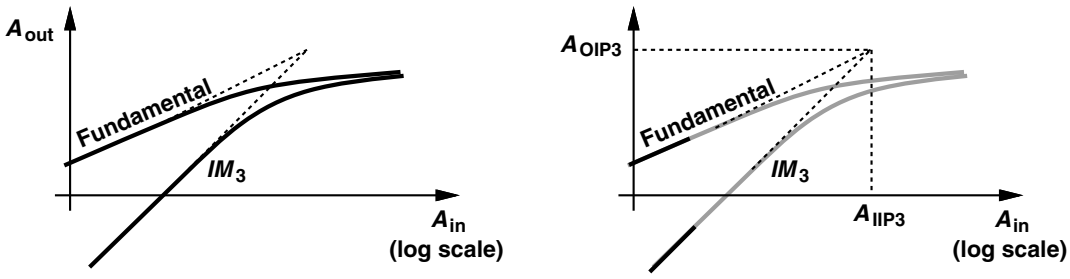
This ratio proves helpful as a sanity check in simulations and measurements.[7] We sometimes write IP_3 rather than IIP_3 if it is clear from the context that the input is of interest.

Upon further consideration, the reader may question the consistency of the above derivations. If the IP_3 is 9.6 dB *higher* than P_{1dB}, is the gain not heavily compressed at $A_{in} = A_{IIP3}$?! If the gain is compressed, why do we still express the amplitude of the fundamentals at the output as $\alpha_1 A$? It appears that we must instead write this amplitude as $[\alpha_1 + (9/4)\alpha_3 A^2]A$ to account for the compression.

In reality, the situation is even more complicated. The value of IP_3 given by Eq. (2.47) may *exceed* the supply voltage, indicating that higher-order nonlinearities manifest themselves as A_{in} approaches A_{IIP3} [Fig. 2.21(a)]. In other words, the IP_3 is not a directly measureable quantity.

In order to avoid these quandaries, we measure the IP_3 as follows. We begin with a very low input level so that $\alpha_1 + (9/4)\alpha_3 A_{in}^2 \approx \alpha_1$ (and, of course, higher order nonlinearities

7. Note that this relationship holds for a third-order system and not necessarily if higher-order terms manifest themselves.



(a) (b)

Figure 2.21 *(a) Actual behavior of nonlinear circuits, (b) definition of IP_3 based on extrapolation.*

are also negligible). We increase A_{in}, plot the amplitudes of the fundamentals and the IM products on a log-log scale, and *extrapolate* these plots according to their slopes (one and three, respectively) to obtain the IP_3 [Fig. 2.21(b)]. To ensure that the signal levels remain well below compression and higher-order terms are negligible, we must observe a 3-dB rise in the IM products for every 1-dB increase in A_{in}. On the other hand, if A_{in} is excessively small, then the output IM components become comparable with the noise floor of the circuit (or the noise floor of the simulated spectrum), thus leading to inaccurate results.

Example 2.11

A low-noise amplifier senses a -80-dBm signal at 2.410 GHz and two -20-dBm interferers at 2.420 GHz and 2.430 GHz. What IIP_3 is required if the IM products must remain 20 dB below the signal? For simplicity, assume 50-Ω interfaces at the input and output.

Solution:

Denoting the peak amplitudes of the signal and the interferers by A_{sig} and A_{int}, respectively, we can write at the LNA output:

$$20\log|\alpha_1 A_{sig}| - 20\,\text{dB} = 20\log\left|\frac{3}{4}\alpha_3 A_{int}^3\right|. \tag{2.50}$$

It follows that

$$|\alpha_1 A_{sig}| = \left|\frac{30}{4}\alpha_3 A_{int}^3\right|. \tag{2.51}$$

In a 50-Ω system, the -80-dBm and -20-dBm levels respectively yield $A_{sig} = 31.6\ \mu V_p$ and $A_{int} = 31.6\,mV_p$. Thus,

$$IIP_3 = \sqrt{\frac{4}{3}\left|\frac{\alpha_1}{\alpha_3}\right|} \tag{2.52}$$

$$= 3.65\,V_p \tag{2.53}$$

$$= +15.2\,\text{dBm}. \tag{2.54}$$

Such an IP_3 is extremely difficult to achieve, especially for a complete receiver chain.

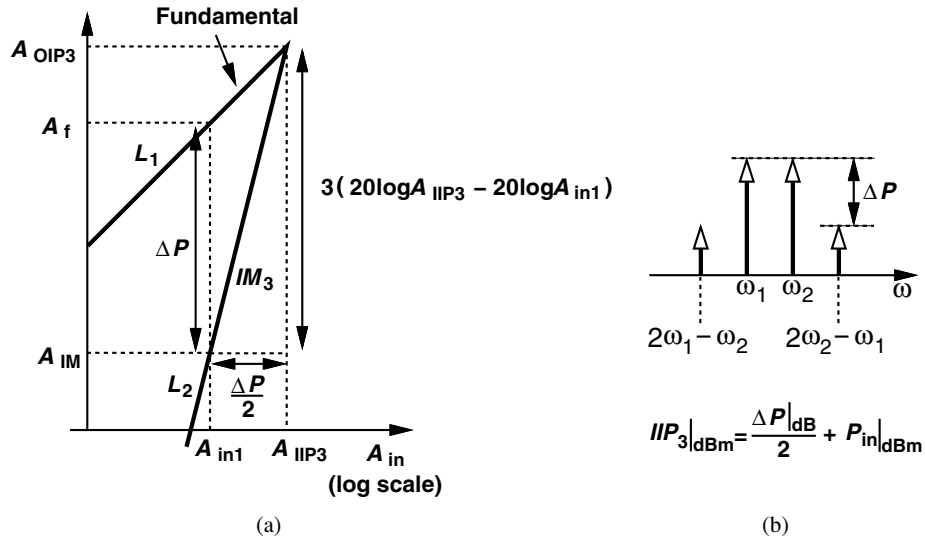


Figure 2.22 *(a) Relationships among various power levels in a two-tone test, (b) illustration of shortcut technique.*

Since extrapolation proves quite tedious in simulations or measurements, we often employ a shortcut that provides a reasonable initial estimate. As illustrated in Fig. 2.22(a), suppose *hypothetically* that the input is equal to A_{IIP3}, and hence the (extrapolated) output IM products are as large as the (extrapolated) fundamental tones. Now, the input is reduced to a level A_{in1}. That is, the change in the input is equal to $20\log A_{IIP3} - 20\log A_{in1}$. On a log-log scale, the IM products fall with a slope of 3 and the fundamentals with a slope of unity. Thus, the *difference* between the two plots increases with a slope of 2. We denote $20\log A_f - 20\log A_{IM}$ by ΔP and write

$$\Delta P = 20\log A_f - 20\log A_{IM} = 2(20\log A_{IIP3} - 20\log A_{in1}), \tag{2.55}$$

obtaining

$$20\log A_{IIP3} = \frac{\Delta P}{2} + 20\log A_{in1}. \tag{2.56}$$

In other words, for a given input level (well below P_{1dB}), the IIP_3 can be calculated by halving the difference between the output fundamental and IM levels and adding the result to the input level, where all values are expressed as logarithmic quantities. Figure 2.22(b) depicts an abbreviated notation for this rule. The key point here is that the IP_3 is measured without extrapolation.

Why do we consider the above result an *estimate*? After all, the derivation assumes third-order nonlinearity. A difficulty arises if the circuit contains *dynamic* nonlinearities, in which case this result may deviate from that obtained by extrapolation. The latter is the standard and accepted method for measuring and reporting the IP_3, but the shortcut method proves useful in understanding the behavior of the device under test.

We should remark that *second-order* nonlinearity also leads to a certain type of intermodulation and is characterized by a "second intercept point," (IP_2).[8] We elaborate on this effect in Chapter 4.

2.2.5 Cascaded Nonlinear Stages

Since in RF systems, signals are processed by cascaded stages, it is important to know how the nonlinearity of each stage is referred to the input of the cascade. The calculation of P_{1dB} for a cascade is outlined in Problem 2.1. Here, we determine the IP_3 of a cascade. For the sake of brevity, we hereafter denote the input IP_3 by A_{IP3} unless otherwise noted.

Consider two nonlinear stages in cascade (Fig. 2.23). If the input/output characteristics of the two stages are expressed, respectively, as

$$y_1(t) = \alpha_1 x(t) + \alpha_2 x^2(t) + \alpha_3 x^3(t) \tag{2.57}$$

$$y_2(t) = \beta_1 y_1(t) + \beta_2 y_1^2(t) + \beta_3 y_1^3(t), \tag{2.58}$$

then

$$\begin{aligned} y_2(t) = {} & \beta_1[\alpha_1 x(t) + \alpha_2 x^2(t) + \alpha_3 x^3(t)] + \beta_2[\alpha_1 x(t) + \alpha_2 x^2(t) + \alpha_3 x^3(t)]^2 \\ & + \beta_3[\alpha_1 x(t) + \alpha_2 x^2(t) + \alpha_3 x^3(t)]^3. \end{aligned} \tag{2.59}$$

Considering only the first- and third-order terms, we have

$$y_2(t) = \alpha_1\beta_1 x(t) + (\alpha_3\beta_1 + 2\alpha_1\alpha_2\beta_2 + \alpha_1^3\beta_3)x^3(t) + \cdots. \tag{2.60}$$

Thus, from Eq. (2.47),

$$A_{IP3} = \sqrt{\frac{4}{3}\left|\frac{\alpha_1\beta_1}{\alpha_3\beta_1 + 2\alpha_1\alpha_2\beta_2 + \alpha_1^3\beta_3}\right|}. \tag{2.61}$$

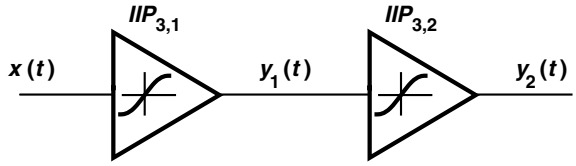


Figure 2.23 *Cascaded nonlinear stages.*

Example 2.12

Two differential pairs are cascaded. Is it possible to select the denominator of Eq. (2.61) such that IP_3 goes to infinity?

(*Continues*)

8. As seen in the next section, second-order nonlinearity also affects the IP_3 in cascaded systems.

Example 2.12 (*Continued*)

Solution:

With no asymmetries in the cascade, $\alpha_2 = \beta_2 = 0$. Thus, we seek the condition $\alpha_3\beta_1 + \alpha_1^3\beta_3 = 0$, or equivalently,

$$\frac{\alpha_3}{\alpha_1} = -\frac{\beta_3}{\beta_1} \cdot \alpha_1^2. \tag{2.62}$$

Since both stages are compressive, $\alpha_3/\alpha_1 < 0$ and $\beta_3/\beta_1 < 0$. It is therefore impossible to achieve an arbitrarily high IP_3.

Equation (2.61) leads to more intuitive results if its two sides are squared and inverted:

$$\frac{1}{A_{IP3}^2} = \frac{3}{4}\left|\frac{\alpha_3\beta_1 + 2\alpha_1\alpha_2\beta_2 + \alpha_1^3\beta_3}{\alpha_1\beta_1}\right| \tag{2.63}$$

$$= \frac{3}{4}\left|\frac{\alpha_3}{\alpha_1} + \frac{2\alpha_2\beta_2}{\beta_1} + \frac{\alpha_1^2\beta_3}{\beta_1}\right| \tag{2.64}$$

$$= \left|\frac{1}{A_{IP3,1}^2} + \frac{3\alpha_2\beta_2}{2\beta_1} + \frac{\alpha_1^2}{A_{IP3,2}^2}\right|, \tag{2.65}$$

where $A_{IP3,1}$ and $A_{IP3,2}$ represent the input IP_3's of the first and second stages, respectively. Note that A_{IP3}, $A_{IP3,1}$, and $A_{IP3,2}$ are voltage quantities.

The key observation in Eq. (2.65) is that to "refer" the IP_3 of the second stage to the input of the cascade, we must divide it by α_1. Thus, the higher the gain of the first stage, the more nonlinearity is contributed by the second stage.

IM Spectra in a Cascade To gain more insight into the above results, let us assume $x(t) = A\cos\omega_1 t + A\cos\omega_2 t$ and identify the IM products in a cascade. With the aid of Fig. 2.24, we make the following observations:[9]

1. The input tones are amplified by a factor of approximately α_1 in the first stage and β_1 in the second. Thus, the output fundamentals are given by $\alpha_1\beta_1 A(\cos\omega_1 t + \cos\omega_2 t)$.
2. The IM products generated by the first stage, namely, $(3\alpha_3/4)A^3[\cos(2\omega_1 - \omega_2)t + \cos(2\omega_2 - \omega_1)t]$, are amplified by a factor of β_1 when they appear at the output of the second stage.
3. Sensing $\alpha_1 A(\cos\omega_1 t + \cos\omega_2 t)$ at its input, the second stage produces its own IM components: $(3\beta_3/4)(\alpha_1 A)^3\cos(2\omega_1 - \omega_2)t + (3\beta_3/4)(\alpha_1 A)^3\cos(2\omega_2 - \omega_1)t$.

9. The spectrum of $A\cos\omega t$ consists of two impulses, each with a weight of $A/2$. We drop the factor of $1/2$ in the figures for simplicity.

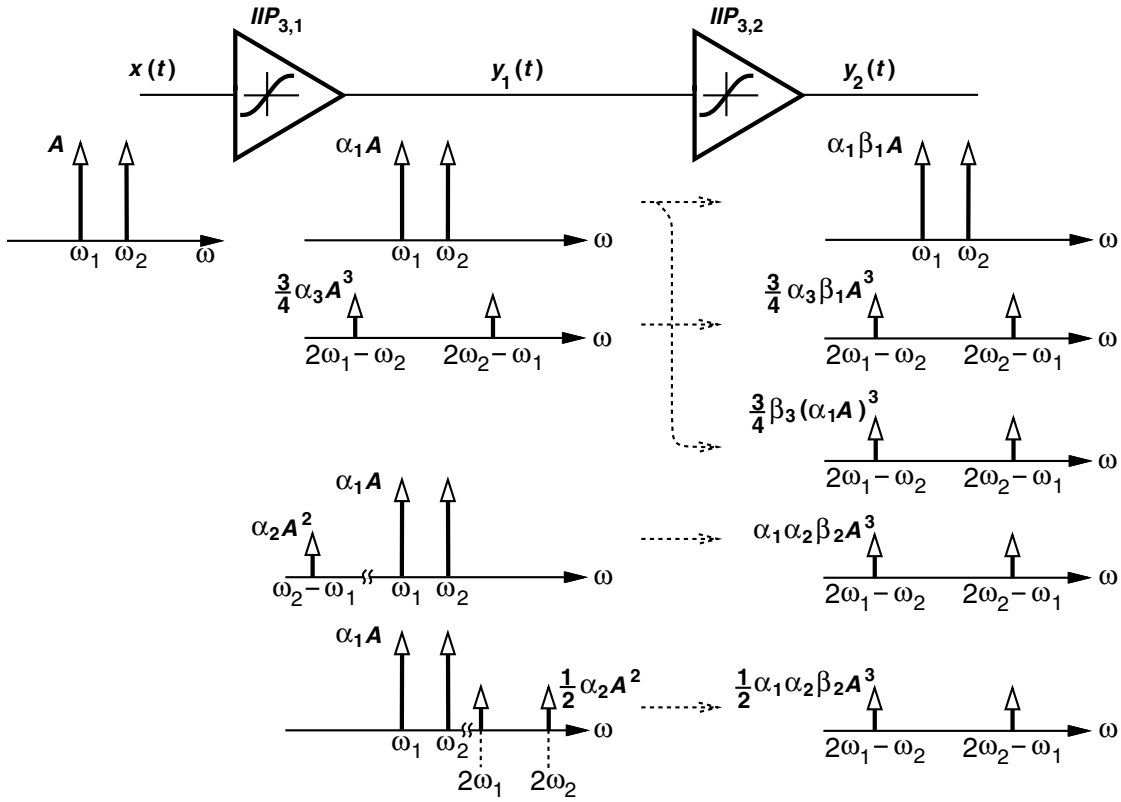


Figure 2.24 *Spectra in a cascade of nonlinear stages.*

4. The second-order nonlinearity in $y_1(t)$ generates components at $\omega_1 - \omega_2$, $2\omega_1$, and $2\omega_2$. Upon experiencing a similar nonlinearity in the second stage, these components are mixed with those at ω_1 and ω_2 and translated to $2\omega_1 - \omega_2$ and $2\omega_2 - \omega_1$. Specifically, as shown in Fig. 2.24, $y_2(t)$ contains terms such as $2\beta_2[\alpha_1 A \cos\omega_1 t \times \alpha_2 A^2 \cos(\omega_1 - \omega_2)t]$ and $2\beta_2(\alpha_1 A \cos\omega_1 t \times 0.5\alpha_2 A^2 \cos 2\omega_2 t)$. The resulting IM products can be expressed as $(3\alpha_1\alpha_2\beta_2 A^3/2)[\cos(2\omega_1 - \omega_2)t + \cos(2\omega_2 - \omega_1)t]$. Interestingly, the cascade of two *second-order* nonlinearities can produce *third-order* IM products.

Adding the amplitudes of the IM products, we have

$$\begin{aligned} y_2(t) = {} & \alpha_1\beta_1 A(\cos\omega_1 t + \cos\omega_2 t) \\ & + \left(\frac{3\alpha_3\beta_1}{4} + \frac{3\alpha_1^3\beta_3}{4} + \frac{3\alpha_1\alpha_2\beta_2}{2}\right) A^3[\cos(\omega_1 - 2\omega_2)t \\ & + \cos(2\omega_2 - \omega_1)t] + \cdots, \end{aligned} \tag{2.66}$$

obtaining the same IP_3 as above. This result assumes zero phase shift for all components.

Why did we add the amplitudes of the IM_3 products in Eq. (2.66) without regard for their phases? Is it possible that phase shifts in the first and second stages allow partial

cancellation of these terms and hence a higher IP_3? Yes, it is possible but uncommon in practice. Since the frequencies ω_1, ω_2, $2\omega_1 - \omega_2$, and $2\omega_2 - \omega_1$ are close to one another, these components experience approximately equal phase shifts.

But how about the terms described in the fourth observation? Components such as $\omega_1 - \omega_2$ and $2\omega_1$ may fall well out of the signal band and experience phase shifts different from those in the first three observations. For this reason, we may consider Eqs. (2.65) and (2.66) as the worst-case scenario. Since most RF systems incorporate narrowband circuits, the terms at $\omega_1 \pm \omega_2$, $2\omega_1$, and $2\omega_2$ are heavily attenuated at the output of the first stage. Consequently, the second term on the right-hand side of (2.65) becomes negligible, and

$$\frac{1}{A_{IP3}^2} \approx \frac{1}{A_{IP3,1}^2} + \frac{\alpha_1^2}{A_{IP3,2}^2}. \tag{2.67}$$

Extending this result to three or more stages, we have

$$\frac{1}{A_{IP3}^2} \approx \frac{1}{A_{IP3,1}^2} + \frac{\alpha_1^2}{A_{IP3,2}^2} + \frac{\alpha_1^2\beta_1^2}{A_{IP3,3}^2} + \cdots. \tag{2.68}$$

Thus, if each stage in a cascade has a gain greater than unity, the nonlinearity of the latter stages becomes increasingly more critical because the IP_3 of each stage is equivalently scaled *down* by the total gain preceding that stage.

Example 2.13

A low-noise amplifier having an input IP_3 of $-10\,\text{dBm}$ and a gain of $20\,\text{dB}$ is followed by a mixer with an input IP_3 of $+4\,\text{dBm}$. Which stage limits the IP_3 of the cascade more? Assume the conceptual picture shown in Fig. 2.1(b) to go between volts and dBm's.

Solution:

With $\alpha_1 = 20\,\text{dB}$, we note that

$$A_{IP3,1} = -10\,\text{dBm} \tag{2.69}$$

$$\frac{A_{IP3,2}}{\alpha_1} = -16\,\text{dBm}. \tag{2.70}$$

Since the scaled IP_3 of the second stage is lower than the IP_3 of the first stage, we say the second stage limits the overall IP_3 more.

In the simulation of a cascade, it is possible to determine which stage limits the linearity more. As depicted in Fig. 2.25, we examine the relative IM magnitudes at the output of each stage (Δ_1 and Δ_2, expressed in dB.) If $\Delta_2 \approx \Delta_1$, the second stage contributes negligible nonlinearity. On the other hand, if Δ_2 is substantially less than Δ_1, then the second stage limits the IP_3.

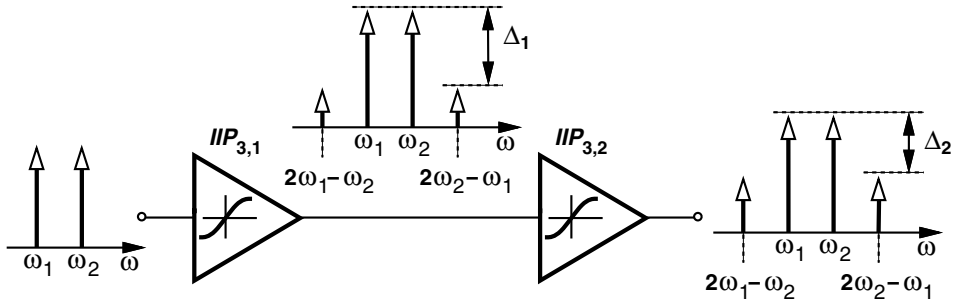


Figure 2.25 *Growth of IM components along the cascade.*

2.2.6 AM/PM Conversion

In some RF circuits, e.g., power amplifiers, amplitude modulation (AM) may be converted to phase modulation (PM), thus producing undesirable effects. In this section, we study this phenomenon.

AM/PM conversion (APC) can be viewed as the dependence of the phase shift upon the signal amplitude. That is, for an input $V_{in}(t) = V_1 \cos \omega_1 t$, the fundamental output component is given by

$$V_{out}(t) = V_2 \cos[\omega_1 t + \phi(V_1)], \tag{2.71}$$

where $\phi(V_1)$ denotes the amplitude-dependent phase shift. This, of course, does not occur in a linear time-invariant system. For example, the phase shift experienced by a sinusoid of frequency ω_1 through a first-order low-pass RC section is given by $-\tan^{-1}(RC\omega_1)$ regardless of the amplitude. Moreover, APC does not appear in a memoryless nonlinear system because the phase shift is zero in this case.

We may therefore surmise that AM/PM conversion arises if a system is both dynamic and nonlinear. For example, if the capacitor in a first-order low-pass RC section is nonlinear, then its "average" value may depend on V_1, resulting in a phase shift, $-\tan^{-1}(RC\omega_1)$, that itself varies with V_1. To explore this point, let us consider the arrangement shown in Fig. 2.26 and assume

$$C_1 = (1 + \alpha V_{out})C_0. \tag{2.72}$$

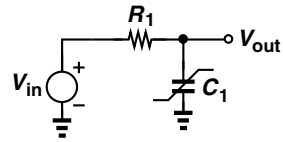


Figure 2.26 *RC section with nonlinear capacitor.*

This capacitor is considered nonlinear because its value depends on its voltage. An exact calculation of the phase shift is difficult here as it requires that we write $V_{in} = R_1 C_1 dV_{out}/dt + V_{out}$ and hence solve

$$V_1 \cos \omega_1 t = R_1(1 + \alpha V_{out})C_0 \frac{dV_{out}}{dt} + V_{out}. \tag{2.73}$$

We therefore make an approximation. Since the value of C_1 varies *periodically* with time, we can express the output as that of a first-order network but with a time-varying

capacitance, $C_1(t)$:

$$V_{out}(t) \approx \frac{V_1}{\sqrt{1 + R_1^2 C_1^2(t)\omega_1^2}} \cos\{\omega_1 t - \tan^{-1}[R_1 C_1(t)\omega_1]\}. \tag{2.74}$$

If $R_1 C_1(t)\omega_1 \ll 1$ rad,

$$V_{out}(t) \approx V_1 \cos[\omega_1 t - R_1(1 + \alpha V_{out})C_0\omega_1]. \tag{2.75}$$

We also assume that $(1 + \alpha V_{out})C_0 \approx (1 + \alpha V_1 \cos\omega_1 t)C_0$, obtaining

$$V_{out}(t) \approx V_1 \cos(\omega_1 t - R_1 C_0\omega_1 - \alpha R_1 C_0\omega_1 V_1 \cos\omega_1 t). \tag{2.76}$$

Does the output *fundamental* contain an input-dependent phase shift here? No, it does not! The reader can show that the third term inside the parentheses produces only higher *harmonics*. Thus, the phase shift of the fundamental is equal to $-R_1 C_0\omega_1$ and hence constant.

The above example entails no AM/PM conversion because of the *first-order* dependence of C_1 upon V_{out}. As illustrated in Fig. 2.27, the average value of C_1 is equal to C_0 regardless of the output amplitude. In general, since C_1 varies periodically, it can be expressed as a Fourier series with a "dc" term representing its average value:

$$C_1(t) = C_{avg} + \sum_{n=1}^{\infty} a_n \cos(n\omega_1 t) + \sum_{n=1}^{\infty} b_n \sin(n\omega_1 t). \tag{2.77}$$

Thus, if C_{avg} is a function of the amplitude, then the phase shift of the fundamental component in the output voltage becomes input-dependent. The following example illustrates this point.

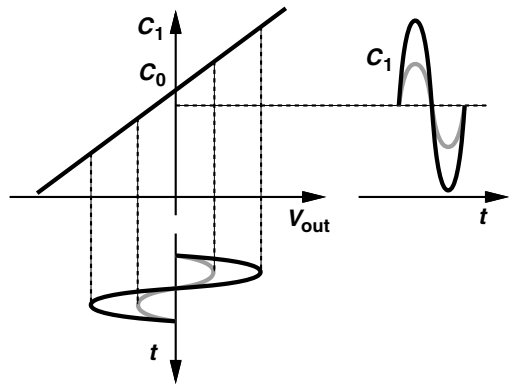


Figure 2.27 *Time variation of capacitor with first-order voltage dependence for small and large swings.*

Example 2.14

Suppose C_1 in Fig. 2.26 is expressed as $C_1 = C_0(1 + \alpha_1 V_{out} + \alpha_2 V_{out}^2)$. Study the AM/PM conversion in this case if $V_{in}(t) = V_1 \cos \omega_1 t$.

Solution:

Figure 2.28 plots $C_1(t)$ for small and large input swings, revealing that C_{avg} indeed depends on the amplitude. We rewrite Eq. (2.75) as

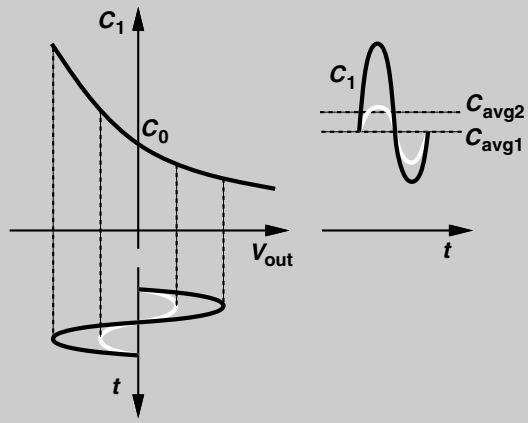


Figure 2.28 *Time variation of capacitor with second-order voltage dependence for small and large swings.*

$$V_{out}(t) \approx V_1 \cos[\omega_1 t - R_1 C_0 \omega_1 (1 + \alpha_1 V_1 \cos \omega_1 t + \alpha_2 V_1^2 \cos^2 \omega_1 t)] \tag{2.78}$$

$$\approx V_1 \cos(\omega_1 t - R_1 C_0 \omega_1 - \frac{\alpha_2 R_1 C_0 \omega_1 V_1^2}{2} - \cdots). \tag{2.79}$$

The phase shift of the fundamental now contains an input-dependent term, $-(\alpha_2 R_1 C_0 \omega_1 V_1^2)/2$. Figure 2.28 also suggests that AM/PM conversion does not occur if the capacitor voltage dependence is odd-symmetric.

What is the effect of APC? In the presence of APC, amplitude modulation (or amplitude noise) corrupts the phase of the signal. For example, if $V_{in}(t) = V_1(1 + m \cos \omega_m t) \cos \omega_1 t$, then Eq. (2.79) yields a phase corruption equal to $-\alpha_2 R_1 C_0 \omega_1 (2mV_1 \cos \omega_m t + m^2 V_1^2 \cos^2 \omega_m t)/2$. We will encounter examples of APC in Chapters 8 and 12.

2.3 NOISE

The performance of RF systems is limited by noise. Without noise, an RF receiver would be able to detect arbitrarily small inputs, allowing communication across arbitrarily long

distances. In this section, we review basic properties of noise and methods of calculating noise in circuits. For a more complete study of noise in analog circuits, the reader is referred to [1].

2.3.1 Noise as a Random Process

The trouble with noise is that it is random. Engineers who are used to dealing with well-defined, deterministic, "hard" facts often find the concept of randomness difficult to grasp, especially if it must be incorporated mathematically. To overcome this fear of randomness, we approach the problem from an intuitive angle.

By "noise is random," we mean the instantaneous value of noise cannot be predicted. For example, consider a resistor tied to a battery and carrying a current [Fig. 2.29(a)]. Due to the ambient temperature, each electron carrying the current experiences thermal agitation, thus following a somewhat random path while, on the average, moving toward the positive terminal of the battery. As a result, the *average* current remains equal to V_B/R but the instantaneous current displays random values.[10]

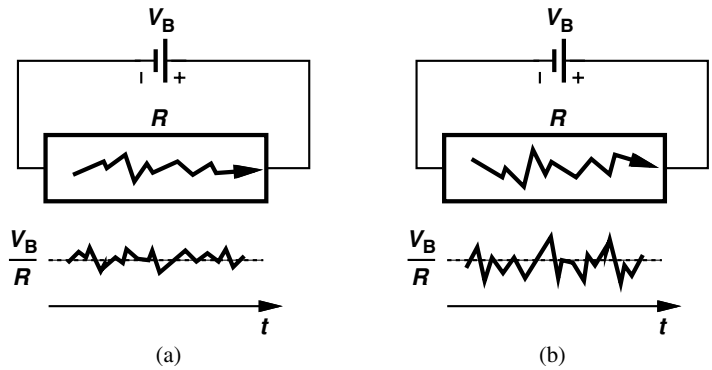


Figure 2.29 *(a) Noise generated in a resistor, (b) effect of higher temperature.*

Since noise cannot be characterized in terms of instantaneous voltages or currents, we seek other attributes of noise that are predictable. For example, we know that a higher ambient temperature leads to greater thermal agitation of electrons and hence larger fluctuations in the current [Fig. 2.29(b)]. How do we express the concept of larger random swings for a current or voltage quantity? This property is revealed by the *average power* of the noise, defined, in analogy with periodic signals, as

$$P_n = \lim_{T\to\infty} \frac{1}{T} \int_0^T n^2(t)dt, \tag{2.80}$$

where $n(t)$ represents the noise waveform. Illustrated in Fig. 2.30, this definition simply means that we compute the area under $n^2(t)$ for a long time, T, and normalize the result to T, thus obtaining the average power. For example, the two scenarios depicted in Fig. 2.29 yield different average powers.

10. As explained later, this is true even with a zero average current.

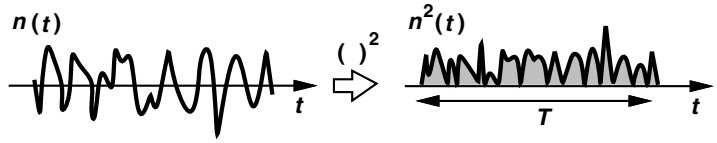


Figure 2.30 *Computation of noise power.*

If $n(t)$ is random, how do we know that P_n is not?! We are fortunate that noise components in circuits have a constant average power. For example, P_n is known and constant for a resistor at a constant ambient temperature.

How long should T in Eq. (2.80) be? Due to its randomness, noise consists of different frequencies. Thus, T must be long enough to accommodate several cycles of the *lowest* frequency. For example, the noise in a crowded restaurant arises from human voice and covers the range of 20 Hz to 20 kHz, requiring that T be on the order of 0.5 s to capture about 10 cycles of the 20-Hz components.[11]

2.3.2 Noise Spectrum

Our foregoing study suggests that the time-domain view of noise provides limited information, e.g., the average power. The frequency-domain view, on the other hand, yields much greater insight and proves more useful in RF design.

The reader may already have some intuitive understanding of the concept of "spectrum." We say the spectrum of human voice spans the range of 20 Hz to 20 kHz. This means that if we somehow measure the frequency content of the voice, we observe all components from 20 Hz to 20 kHz. How, then, do we measure a signal's frequency content, e.g., the strength of a component at 10 kHz? We would need to filter out the remainder of the spectrum and measure the *average power* of the 10-kHz component. Figure 2.31(a) conceptually illustrates such an experiment, where the microphone signal is applied to a band-pass filter having a 1-Hz bandwidth centered around 10 kHz. If a person speaks into the microphone at a steady volume, the power meter reads a constant value.

The scheme shown in Fig. 2.31(a) can be readily extended so as to measure the strength of all frequency components. As depicted in Fig. 2.31(b), a bank of 1-Hz band-pass filters centered at $f_1 \cdots f_n$ measures the average power at each frequency.[12] Called the spectrum or the "power spectral density" (PSD) of $x(t)$ and denoted by $S_x(f)$, the resulting plot displays the average power that the voice (or the noise) carries in a 1-Hz bandwidth at different frequencies.[13]

It is interesting to note that the total area under $S_x(f)$ represents the average power carried by $x(t)$:

$$\int_0^\infty S_x(f)df = \lim_{T\to\infty} \frac{1}{T} \int_0^T x^2(t)dt. \tag{2.81}$$

11. In practice, we make a guess for T, calculate P_n, increase T, recalculate P_n, and repeat until consecutive values of P_n become nearly equal.

12. This is also the conceptual operation of spectrum analyzers.

13. In the theory of signals and systems, the PSD is defined as the Fourier transform of the autocorrelation of a signal. These two views are equivalent.

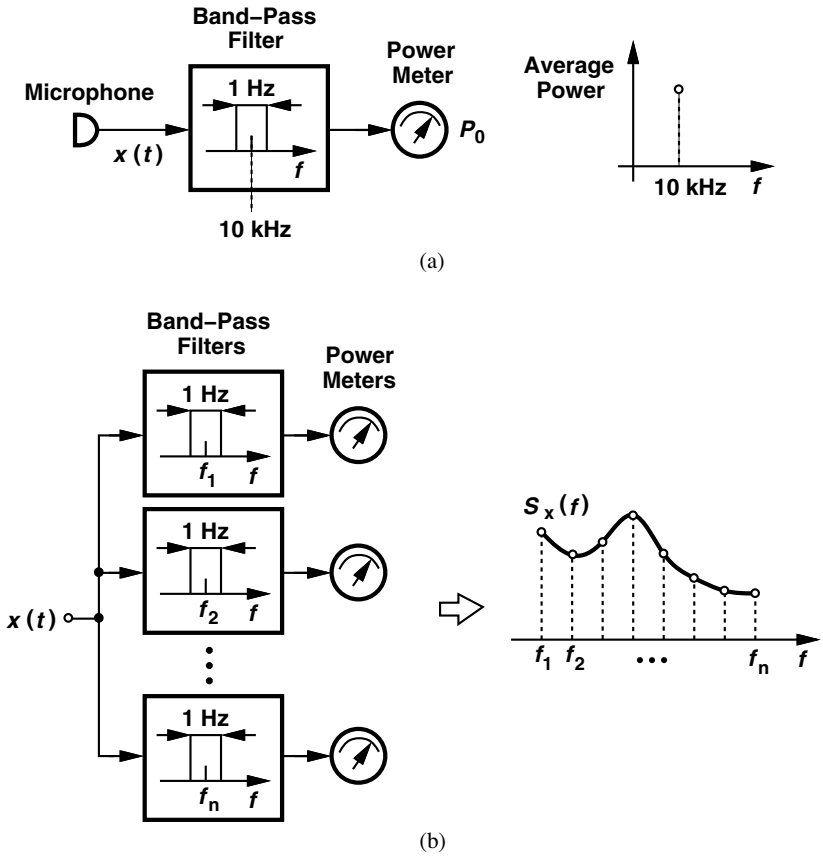


Figure 2.31 *Measurement of (a) power in 1 Hz, and (b) the spectrum.*

The spectrum shown in Fig. 2.31(b) is called "one-sided" because it is constructed for positive frequencies. In some cases, the analysis is simpler if a "two-sided" spectrum is utilized. The latter is an even-symmetric of the former scaled down vertically by a factor of two (Fig. 2.32), so that the two carry equal energies.

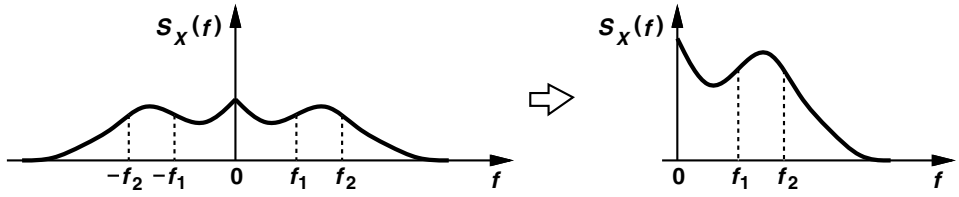


Figure 2.32 *Two-sided and one-sided spectra.*

Example 2.15

A resistor of value R_1 generates a noise voltage whose one-sided PSD is given by

$$S_v(f) = 4kTR_1, \tag{2.82}$$

Example 2.15 (*Continued*)

where $k = 1.38 \times 10^{-23}$ J/K denotes the Boltzmann constant and T the absolute temperature. Such a flat PSD is called "white" because, like white light, it contains all frequencies with equal power levels.

(a) What is the total average power carried by the noise voltage?

(b) What is the dimension of $S_v(f)$?

(c) Calculate the noise voltage for a 50-Ω resistor in 1 Hz at room temperature.

Solution:

(a) The area under $S_v(f)$ appears to be infinite, an implausible result because the resistor noise arises from the finite ambient heat. In reality, $S_v(f)$ begins to fall at $f > 1$ THz, exhibiting a finite total energy, i.e., thermal noise is not quite white.

(b) The dimension of $S_v(f)$ is voltage squared per unit bandwidth (V^2/Hz) rather than power per unit bandwidth (W/Hz). In fact, we may write the PSD as

$$\overline{V_n^2} = 4kTR, \tag{2.83}$$

where $\overline{V_n^2}$ denotes the average power of V_n in 1 Hz.[14] While some texts express the right-hand side as $4kTR\Delta f$ to indicate the total noise in a bandwidth of Δf, we omit Δf with the understanding that our PSDs always represent power in 1 Hz. We shall use $S_v(f)$ and $\overline{V_n^2}$ interchangeably.

(c) For a 50-Ω resistor at $T = 300$ K,

$$\overline{V_n^2} = 8.28 \times 10^{-19}\ \text{V}^2/\text{Hz}. \tag{2.84}$$

This means that if the noise voltage of the resistor is applied to a 1-Hz band-pass filter centered at any frequency (< 1 THz), then the average measured output is given by the above value. To express the result as a root-mean-squared (rms) quantity and in more familiar units, we may take the square root of both sides:

$$\sqrt{\overline{V_n^2}} = 0.91\ \text{nV}/\sqrt{\text{Hz}}. \tag{2.85}$$

The familiar unit is nV but the strange unit is $\sqrt{\text{Hz}}$. The latter bears no profound meaning; it simply says that the average power in 1 Hz is $(0.91\ \text{nV})^2$.

2.3.3 Effect of Transfer Function on Noise

The principal reason for defining the PSD is that it allows many of the frequency-domain operations used with deterministic signals to be applied to random signals as well. For

14. Also called "spot noise."

example, if white noise is applied to a low-pass filter, how do we determine the PSD at the output? As shown in Fig. 2.33, we intuitively expect that the output PSD assumes the shape of the filter's frequency response. In fact, if $x(t)$ is applied to a linear, time-invariant system with a transfer function $H(s)$, then the output spectrum is

$$S_y(f) = S_x(f)|H(f)|^2, \tag{2.86}$$

where $H(f) = H(s = j2\pi f)$ [2]. We note that $|H(f)|$ is squared because $S_x(f)$ is a (voltage or current) squared quantity.

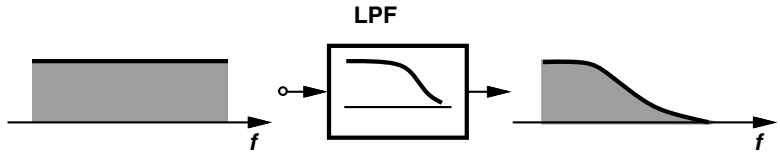


Figure 2.33 *Effect of low-pass filter on white noise.*

2.3.4 Device Noise

In order to analyze the noise performance of circuits, we wish to model the noise of their constituent elements by familiar components such as voltage and current sources. Such a representation allows the use of standard circuit analysis techniques.

Thermal Noise of Resistors As mentioned previously, the ambient thermal energy leads to random agitation of charge carriers in resistors and hence noise. The noise can be modeled by a series voltage source with a PSD of $\overline{V_n^2} = 4kTR_1$ [Thevenin equivalent, Fig. 2.34(a)] or a parallel current source with a PSD of $\overline{I_n^2} = \overline{V_n^2}/R_1 = 4kT/R_1$ [Norton equivalent, Fig. 2.34(b)]. The choice of the model sometimes simplifies the analysis. The polarity of the sources is unimportant (but must be kept the same throughout the calculations of a given circuit).

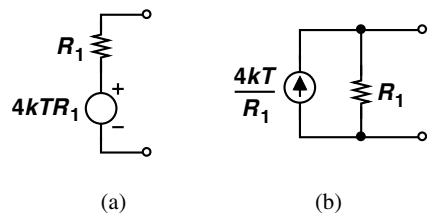


Figure 2.34 *(a) Thevenin and (b) Norton models of resistor thermal noise.*

Example 2.16

Sketch the PSD of the noise voltage measured across the parallel RLC tank depicted in Fig. 2.35(a).

Example 2.16 (*Continued*)

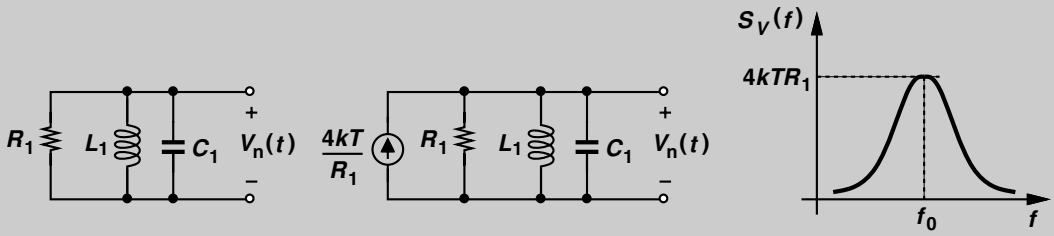


Figure 2.35 *(a) RLC tank, (b) inclusion of resistor noise, (c) output noise spectrum due to* R_1.

Solution:

Modeling the noise of R_1 by a current source, $\overline{I_{n1}^2} = 4kT/R_1$, [Fig. 2.35(b)] and noting that the transfer function V_n/I_{n1} is, in fact, equal to the impedance of the tank, Z_T, we write from Eq. (2.86)

$$\overline{V_n^2} = \overline{I_{n1}^2}|Z_T|^2. \tag{2.87}$$

At $f_0 = (2\pi\sqrt{L_1C_1})^{-1}$, L_1 and C_1 resonate, reducing the circuit to only R_1. Thus, the output noise at f_0 is simply equal to $\overline{I_{n1}^2}R_1^2 = 4kTR_1$. At lower or higher frequencies, the impedance of the tank falls and so does the output noise [Fig. 2.35(c)].

If a resistor converts the ambient heat to a noise voltage or current, can we extract energy from the resistor? In particular, does the arrangement shown in Fig. 2.36 deliver energy to R_2? Interestingly, if R_1 and R_2 reside at the same temperature, no net energy is transferred between them because R_2 also produces a noise PSD of $4kTR_2$ (Problem 2.8). However, suppose R_2 is held at $T = 0\,\text{K}$. Then, R_1 continues to draw thermal energy from its environment, converting it to noise and delivering the energy to R_2. The average power transferred to R_2 is equal to

$$P_{R2} = \frac{\overline{V_{out}^2}}{R_2} \tag{2.88}$$

$$= \overline{V_n^2}\left(\frac{R_2}{R_1 + R_2}\right)^2 \frac{1}{R_2} \tag{2.89}$$

$$= 4kT\frac{R_1R_2}{(R_1 + R_2)^2}. \tag{2.90}$$

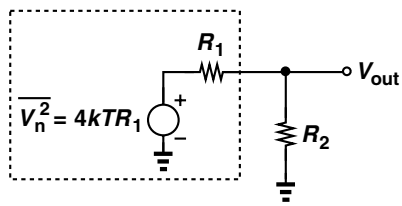


Figure 2.36 *Transfer of noise from one resistor to another.*

This quantity reaches a maximum if $R_2 = R_1$:

$$P_{R2,max} = kT. \tag{2.91}$$

Called the "available noise power," kT is independent of the resistor value and has the dimension of *power* per unit bandwidth. The reader can prove that $kT = -173.8$ dBm/Hz at $T = 300$ K.

For a circuit to exhibit a thermal noise density of $\overline{V_n^2} = 4kTR_1$, it need not contain an explicit resistor of value R_1. After all, Eq. (2.86) suggests that the noise density of a resistor may be transformed to a higher or lower value by the surrounding circuit. We also note that if a passive circuit *dissipates* energy, then it must contain a physical resistance[15] and must therefore *produce* thermal noise. We loosely say "lossy circuits are noisy."

A theorem that consolidates the above observations is as follows: If the real part of the impedance seen between two terminals of a passive (reciprocal) network is equal to $Re\{Z_{out}\}$, then the PSD of the thermal noise seen between these terminals is given by $\overline{V_n^2} = 4kTRe\{Z_{out}\}$ (Fig. 2.37) [8]. This general theorem is not limited to lumped circuits. For example, consider a transmitting antenna that dissipates energy by radiation according to the equation $V_{TX,rms}^2/R_{rad}$, where R_{rad} is the "radiation resistance" [Fig. 2.38(a)]. As a receiving element [Fig. 2.38(b)], the antenna *generates* a thermal noise PSD of[16]

$$\overline{V_{n,ant}^2} = 4kTR_{rad}. \tag{2.92}$$

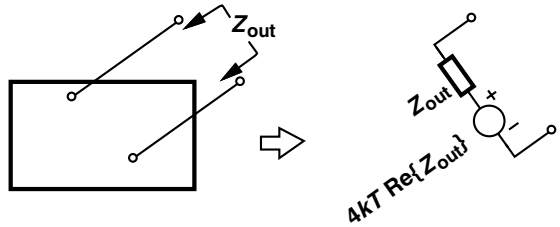
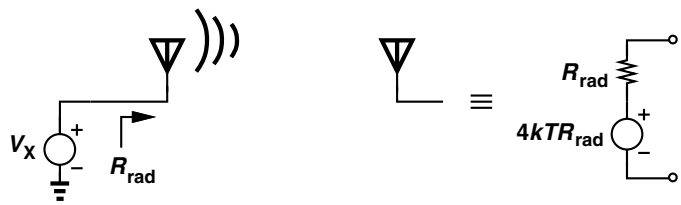


Figure 2.37 *Output noise of a passive (reciprocal) circuit.*

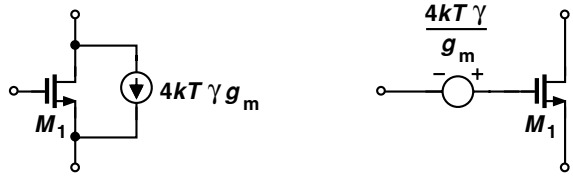


(a) (b)

Figure 2.38 *(a) Transmitting antenna, (b) receiving antenna producing thermal noise.*

15. Recall that ideal inductors and capacitors *store* energy but do not dissipate it.

16. Strictly speaking, this is not correct because the noise of a receiving antenna is in fact given by the "background" noise (e.g., cosmic radiation). However, in RF design, the antenna noise is commonly assumed to be $4kTR_{rad}$.



(a) (b)

Figure 2.39 *Thermal channel noise of a MOSFET modeled as a (a) current source, (b) voltage source.*

Noise in MOSFETs The thermal noise of MOS transistors operating in the saturation region is approximated by a current source tied between the source and drain terminals [Fig. 2.39(a)]:

$$\overline{I_n^2} = 4kT\gamma g_m, \tag{2.93}$$

where γ is the "excess noise coefficient" and g_m the transconductance.[17] The value of γ is $2/3$ for long-channel transistors and may rise to even 2 in short-channel devices [4]. The actual value of γ has other dependencies [5] and is usually obtained by measurements for each generation of CMOS technology. In Problem 2.10, we prove that the noise can alternatively be modeled by a voltage source $\overline{V_n^2} = 4kT\gamma/g_m$ in series with the gate [Fig. 2.39(b)].

Another component of thermal noise arises from the gate resistance of MOSFETs, an effect that becomes increasingly more important as the gate length is scaled down. Illustrated in Fig. 2.40(a) for a device with a width of W and a length of L, this resistance amounts to

$$R_G = \frac{W}{L} R_\square, \tag{2.94}$$

where $R_\square$ denotes the sheet resistance (resistance of one square) of the polysilicon gate. For example, if $W = 1\ \mu$m, $L = 45$ nm, and $R_\square = 15\ \Omega$, then $R_G = 333\ \Omega$. Since R_G is *distributed* over the width of the transistor [Fig. 2.40(b)], its noise must be calculated carefully. As proved in [6], the structure can be reduced to a lumped model having an equivalent gate resistance of $R_G/3$ with a thermal noise PSD of $4kTR_G/3$ [Fig. 2.40(c)]. In a good design, this noise must be much less than that of the channel:

$$4kT\frac{R_G}{3} \ll \frac{4kT\gamma}{g_m}. \tag{2.95}$$

The gate and drain terminals also exhibit physical resistances, which are minimized through the use of multiple fingers.

At very high frequencies the thermal noise current flowing through the channel couples to the gate capacitively, thus generating a "gate-induced noise current" [3] (Fig. 2.41). This

17. More accurately, $\overline{I_n^2} = 4kT\gamma g_{d0}$, where g_{d0} is the drain-source conductance in the triode region (even though the noise is measured in saturation) [3].

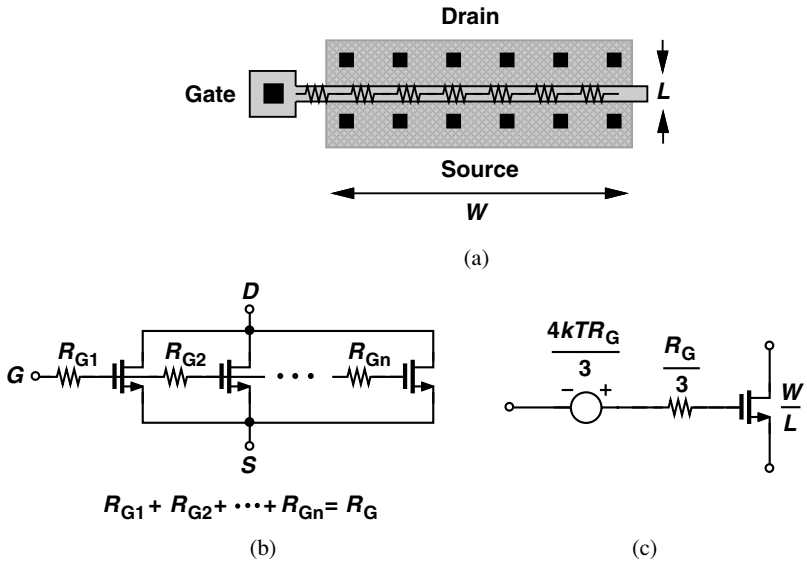


Figure 2.40 *(a) Gate resistance of a MOSFET, (b) equivalent circuit for noise calculation, (c) equivalent noise and resistance in lumped model.*

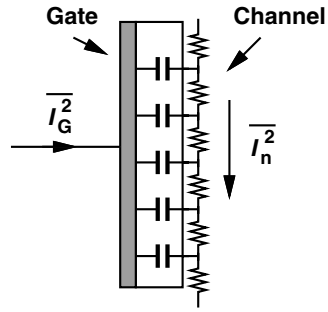


Figure 2.41 *Gate-induced noise,* $\overline{I_G^2}$.

effect is not modeled in typical circuit simulators, but its significance has remained unclear. In this book, we neglect the gate-induced noise current.

MOS devices also suffer from "flicker" or "$1/f$" noise. Modeled by a voltage source in series with the gate, this noise exhibits the following PSD:

$$\overline{V_n^2} = \frac{K}{WLC_{ox}} \frac{1}{f}, \tag{2.96}$$

where K is a process-dependent constant. In most CMOS technologies, K is lower for PMOS devices than for NMOS transistors because the former carry charge well below the silicon-oxide interface and hence suffer less from "surface states" (dangling bonds) [1]. The $1/f$ dependence means that noise components that vary slowly assume a large amplitude. The choice of the lowest frequency in the noise integration depends on the time scale of interest and/or the spectrum of the desired signal [1].

Example 2.17

Can the flicker noise be modeled by a current source?

Solution:

Yes, as shown in Fig. 2.42, a MOSFET having a small-signal voltage source of magnitude V_1 in series with its gate is equivalent to a device with a current source of value $g_m V_1$ tied between drain and source. Thus,

$$\overline{I_1^2} = g_m^2 \frac{K}{WLC_{ox}} \frac{1}{f}. \tag{2.97}$$

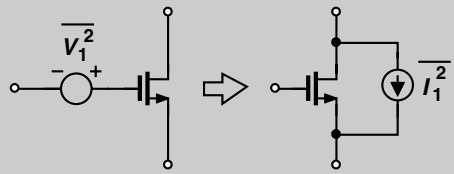


Figure 2.42 *Conversion of flicker noise voltage to current.*

For a given device size and bias current, the $1/f$ noise PSD intercepts the thermal noise PSD at some frequency, called the "$1/f$ noise corner frequency," f_c. Illustrated in Fig. 2.43, f_c can be obtained by converting the flicker noise voltage to current (according to the above example) and equating the result to the thermal noise current:

$$\frac{K}{WLC_{ox}} \frac{1}{f_c} g_m^2 = 4kT\gamma g_m. \tag{2.98}$$

It follows that

$$f_c = \frac{K}{WLC_{ox}} \frac{g_m}{4kT\gamma}. \tag{2.99}$$

The corner frequency falls in the range of tens or even hundreds of megahertz in today's MOS technologies.

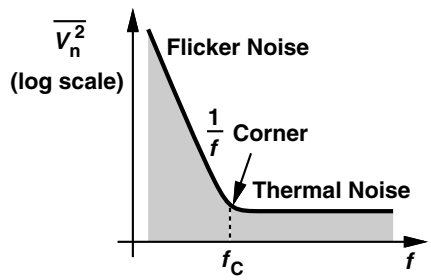


Figure 2.43 *Flicker noise corner frequency.*

While the effect of flicker noise may seem negligible at high frequencies, we must note that nonlinearity or time variance in circuits such as mixers and oscillators may translate the $1/f$-shaped spectrum to the RF range. We study these phenomena in Chapters 6 and 8.

Noise in Bipolar Transistors Bipolar transistors contain physical resistances in their base, emitter, and collector regions, all of which generate thermal noise. Moreover, they also suffer from "shot noise" associated with the transport of carriers across the base-emitter junction. As shown in Fig. 2.44, this noise is modeled by two current sources having the following PSDs:

$$\overline{I_{n,b}^2} = 2qI_B = 2q\frac{I_C}{\beta} \tag{2.100}$$

$$\overline{I_{n,c}^2} = 2qI_C, \tag{2.101}$$

where I_B and I_C are the base and collector bias currents, respectively. Since $g_m = I_C/(kT/q)$ for bipolar transistors, the collector current shot noise is often expressed as

$$\overline{I_{n,c}^2} = 4kT\frac{g_m}{2}, \tag{2.102}$$

in analogy with the thermal noise of MOSFETs or resistors.

In low-noise circuits, the base resistance thermal noise and the collector current shot noise become dominant. For this reason, wide transistors biased at high current levels are employed.

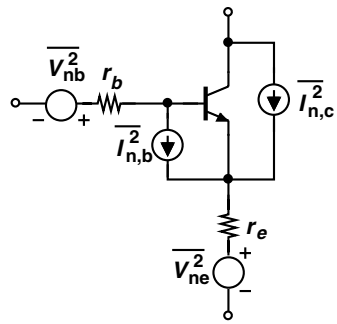


Figure 2.44 *Noise sources in a bipolar transistor.*

2.3.5 Representation of Noise in Circuits

With the noise of devices formulated above, we now wish to develop *measures* of the noise performance of circuits, i.e., metrics that reveal how noisy a given circuit is.

Input-Referred Noise How can the noise of a circuit be observed in the laboratory? We have access only to the output and hence can measure only the output noise. Unfortunately, the output noise does not permit a fair comparison between circuits: a circuit may exhibit high output noise because it has a high gain rather than high noise. For this reason, we "refer" the noise to the input.

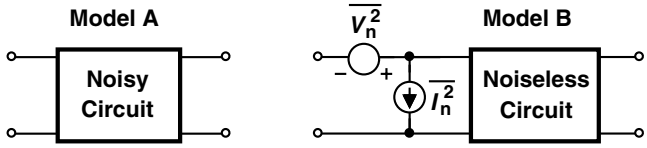


Figure 2.45 *Input-referred noise.*

In analog design, the input-referred noise is modeled by a series voltage source and a parallel current source (Fig. 2.45) [1]. The former is obtained by shorting the input port of models A and B and equating their output noises (or, equivalently, dividing the output noise by the voltage gain). Similarly, the latter is computed by leaving the input ports open and equating the output noises (or, equivalently, dividing the output noise by the transimpedance gain).

Example 2.18

Calculate the input-referred noise of the common-gate stage depicted in Fig. 2.46(a). Assume I_1 is ideal and neglect the noise of R_1.

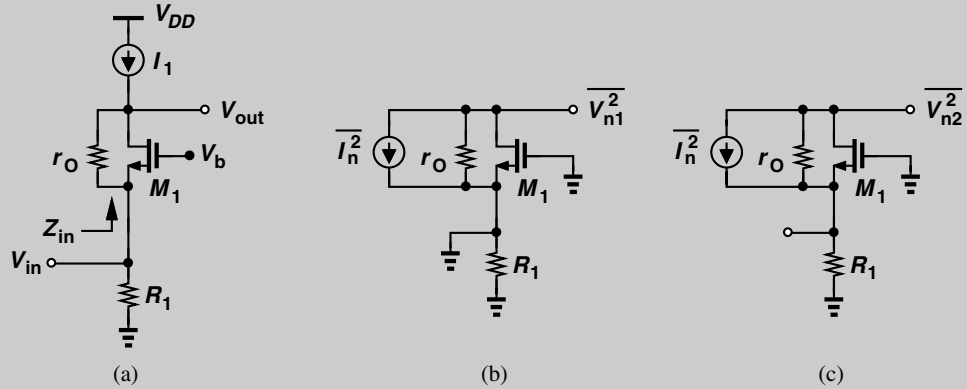


Figure 2.46 *(a) CG stage, (b) computation of input-referred noise voltage, (c) computation of input-referred noise current.*

Solution:

Shorting the input to ground, we write from Fig. 2.46(b),

$$\overline{V_{n1}^2} = \overline{I_n^2} \cdot r_O^2. \tag{2.103}$$

Since the voltage gain of the stage is given by $1 + g_m r_O$, the input-referred noise voltage is equal to

$$\overline{V_{n,in}^2} = \frac{\overline{I_n^2} r_O^2}{(1 + g_m r_O)^2} \tag{2.104}$$

$$\approx \frac{4kT\gamma}{g_m}, \tag{2.105}$$

(*Continues*)

Example 2.18 (*Continued*)

where it is assumed $g_m r_O \gg 1$. Leaving the input open as shown in Fig. 2.46(c), the reader can show that (Problem 2.12)

$$\overline{V_{n2}^2} = \overline{I_n^2} r_O^2. \tag{2.106}$$

Defined as the output voltage divided by the input current, the transimpedance gain of the stage is given by $g_m r_O R_1$ (why?). It follows that

$$\overline{I_{n,in}^2} = \frac{\overline{I_n^2} r_O^2}{g_m^2 r_O^2 R_1^2} \tag{2.107}$$

$$= \frac{4kT\gamma}{g_m R_1^2}. \tag{2.108}$$

From the above example, it may appear that the noise of M_1 is "counted" twice. It can be shown that [1] the two input-referred noise sources are necessary and sufficient, but often correlated.

Example 2.19

Explain why the output noise of a circuit depends on the output impedance of the *preceding* stage.

Solution:

Modeling the noise of the circuit by input-referred sources as shown in Fig. 2.47, we observe that some of $\overline{I_n^2}$ flows through Z_1, generating a noise voltage at the input that depends on $|Z_1|$. Thus, the output noise, $V_{n,out}$, also depends on $|Z_1|$.

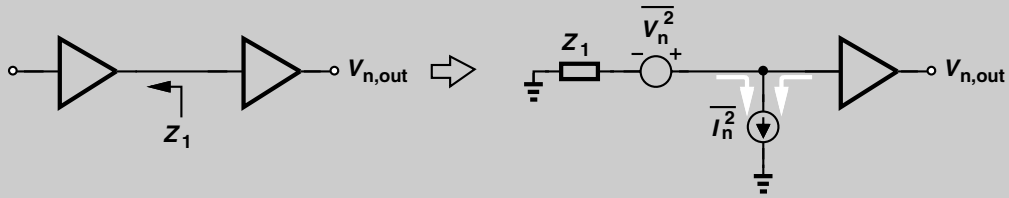


Figure 2.47 *Noise in a cascade.*

The computation and use of input-referred noise sources prove difficult at high frequencies. For example, it is quite challenging to measure the transimpedance gain of an RF stage. For this reason, RF designers employ the concept of "noise figure" as another metric of noise performance that more easily lends itself to measurement.

Noise Figure In circuit and system design, we are interested in the signal-to-noise ratio (SNR), defined as the signal power divided by the noise power. It is therefore helpful to

ask, how does the SNR degrade as the signal travels through a given circuit? If the circuit contains no noise, then the output SNR is *equal* to the input SNR even if the circuit acts as an attenuator.[18] To quantify how noisy the circuit is, we define its noise figure (NF) as

$$\text{NF} = \frac{SNR_{in}}{SNR_{out}} \tag{2.109}$$

such that it is equal to 1 for a noiseless stage. Since each quantity in this ratio has a dimension of power (or voltage squared), we express NF in decibels as

$$\text{NF}|_{\text{dB}} = 10 \log \frac{SNR_{in}}{SNR_{out}}. \tag{2.110}$$

Note that most texts call (2.109) the "noise factor" and (2.110) the noise figure. We do not make this distinction in this book.

Compared to input-referred noise, the definition of NF in (2.109) may appear rather complicated: it depends on not only the noise of the circuit under consideration but the SNR provided by the *preceding stage*. In fact, if the input signal contains no noise, then $SNR_{in} = \infty$ and $\text{NF} = \infty$, even though the circuit may have finite internal noise. For such a case, NF is not a meaningful parameter and only the input-referred noise can be specified.

Calculation of the noise figure is generally simpler than Eq. (2.109) may suggest. For example, suppose a low-noise amplifier senses the signal received by an antenna [Fig. 2.48(a)]. As predicted by Eq. (2.92), the antenna "radiation resistance," R_S, produces thermal noise, leading to the model shown in Fig. 2.48(b). Here, $\overline{V_{n,RS}^2}$ represents the thermal noise of the antenna, and $\overline{V_n^2}$ the output noise of the LNA. We must compute SNR_{in} at the LNA input and SNR_{out} at its output.

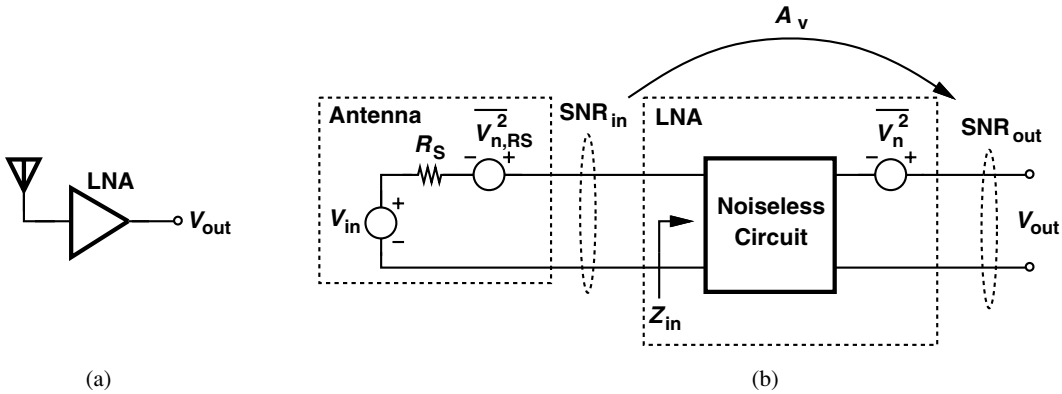


Figure 2.48 *(a) Antenna followed by LNA, (b) equivalent circuit.*

18. Because the input signal and the input noise are attenuated by the same factor.

If the LNA exhibits an input impedance of Z_{in}, then both V_{in} and V_{RS} experience an attenuation factor of $\alpha = Z_{in}/(Z_{in} + R_S)$ as they appear at the input of the LNA. That is,

$$SNR_{in} = \frac{|\alpha|^2 V_{in}^2}{|\alpha|^2 \overline{V_{RS}^2}}, \tag{2.111}$$

where V_{in} denotes the rms value of the signal received by the antenna.

To determine SNR_{out}, we assume a voltage gain of A_v from the LNA input to the output and recognize that the output signal power is equal to $V_{in}^2|\alpha|^2A_v^2$. The output noise consists of two components: (a) the noise of the antenna amplified by the LNA, $\overline{V_{RS}^2}|\alpha|^2A_v^2$, and (b) the output noise of the LNA, $\overline{V_n^2}$. Since these two components are uncorrelated, we simply add the PSDs and write

$$SNR_{out} = \frac{V_{in}^2|\alpha|^2A_v^2}{\overline{V_{RS}^2}|\alpha|^2A_v^2 + \overline{V_n^2}}. \tag{2.112}$$

It follows that

$$\text{NF} = \frac{V_{in}^2}{4kTR_S} \cdot \frac{\overline{V_{RS}^2}|\alpha|^2A_v^2 + \overline{V_n^2}}{V_{in}^2|\alpha|^2A_v^2} \tag{2.113}$$

$$= \frac{1}{\overline{V_{RS}^2}} \cdot \frac{\overline{V_{RS}^2}|\alpha|^2A_v^2 + \overline{V_n^2}}{|\alpha|^2A_v^2} \tag{2.114}$$

$$= 1 + \frac{\overline{V_n^2}}{|\alpha|^2A_v^2} \cdot \frac{1}{\overline{V_{RS}^2}}. \tag{2.115}$$

This result leads to another definition of the NF: the total noise at the output divided by the noise at the output due to the source impedance. The NF is usually specified for a 1-Hz bandwidth at a given frequency, and hence sometimes called the "spot noise figure" to emphasize the small bandwidth.

Equation (2.115) suggests that the NF depends on the *source impedance*, not only through $\overline{V_{RS}^2}$ but also through $\overline{V_n^2}$ (Example 2.19). In fact, if we model the noise by *input-referred* sources, then the input noise current, $\overline{I_{n,in}^2}$, partially flows through R_S, generating a source-dependent noise voltage of $\overline{I_{n,in}^2}R_S^2$ at the input and hence a proportional noise at the output. Thus, the NF must be specified with respect to a source impedance—typically 50 Ω.

For hand analysis and simulations, it is possible to reduce the right-hand side of Eq. (2.114) to a simpler form by noting that the numerator is the *total* noise measured at the output:

$$\text{NF} = \frac{1}{4kTR_S} \cdot \frac{\overline{V_{n,out}^2}}{A_0^2}, \tag{2.116}$$

where $\overline{V_{n,out}^2}$ includes both the source impedance noise and the LNA noise, and $A_0 = |\alpha|A_v$ is the voltage gain from V_{in} to V_{out} (rather than the gain from the LNA input to its output). We loosely say, "to calculate the NF, we simply divide the total output noise by the gain

from V_{in} to V_{out} and normalize the result to the noise of R_S." Alternatively, we can say from (2.115) that "we calculate the output noise due to the amplifier ($\overline{V_n^2}$), divide it by the gain, normalize it to $4kTR_S$, and add 1 to the result."

It is important to note that the above derivations are valid even if no actual *power* is transferred from the antenna to the LNA or from the LNA to a load. For example, if Z_{in} in Fig. 2.48(b) goes to infinity, no power is delivered to the LNA, but all of the derivations remain valid because they are based on *voltage* (squared) quantities rather than power quantities. In other words, so long as the derivations incorporate noise and signal voltages, no inconsistency arises in the presence of impedance mismatches or even infinite input impedances. This is a critical difference in thinking between modern RF design and traditional microwave design.

Example 2.20

Compute the noise figure of a shunt resistor R_P with respect to a source impedance R_S [Fig. 2.49(a)].

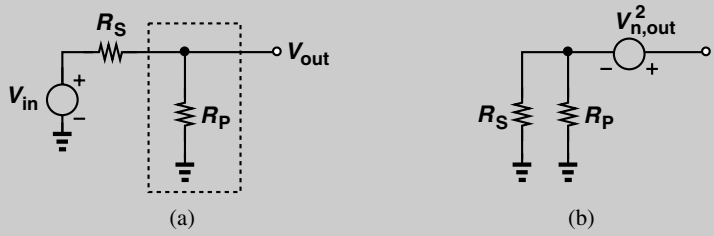


Figure 2.49 *(a) Circuit consisting of a single parallel resistor, (b) model for NF calculation.*

Solution:

From Fig. 2.49(b), the total output noise voltage is obtained by setting V_{in} to zero:

$$\overline{V_{n,out}^2} = 4kT(R_S||R_P). \tag{2.117}$$

The gain is equal to

$$A_0 = \frac{R_P}{R_P + R_S}. \tag{2.118}$$

Thus,

$$\text{NF} = 4kT(R_S||R_P)\frac{(R_S + R_P)^2}{R_P^2}\frac{1}{4kTR_S} \tag{2.119}$$

$$= 1 + \frac{R_S}{R_P}. \tag{2.120}$$

The NF is therefore minimized by *maximizing* R_P. Note that if $R_P = R_S$ to provide impedance matching, then the NF cannot be less than 3 dB. We will return to this critical point in the context of LNA design in Chapter 5.

Example 2.21

Determine the noise figure of the common-source stage shown in Fig. 2.50(a) with respect to a source impedance R_S. Neglect the capacitances and flicker noise of M_1 and assume I_1 is ideal.

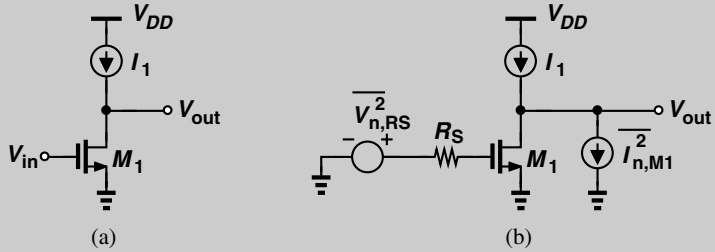


Figure 2.50 *(a) CS stage, (b) inclusion of noise.*

Solution:

From Fig. 2.50(b), the output noise consists of two components: (a) that due to M_1, $\overline{I_{n,M1}^2} r_O^2$, and (b) the amplified noise of R_S, $\overline{V_{RS}^2}(g_m r_O)^2$. It follows that

$$\text{NF} = \frac{4kT\gamma g_m r_O^2 + 4kTR_S(g_m r_O)^2}{(g_m r_O)^2} \cdot \frac{1}{4kTR_S} \tag{2.121}$$

$$= \frac{\gamma}{g_m R_S} + 1. \tag{2.122}$$

This result implies that the NF falls as R_S rises. Does this mean that, even though the amplifier remains unchanged, the overall system noise performance *improves* as R_S increases?! This interesting point is studied in Problems 2.18 and 2.19.

Noise Figure of Cascaded Stages Since many stages appear in a receiver chain, it is desirable to determine the NF of the overall cascade in terms of that of each stage. Consider the cascade depicted in Fig. 2.51(a), where A_{v1} and A_{v2} denote the *unloaded* voltage gain of the two stages. The input and output impedances and the output noise voltages of the two stages are also shown.[19]

We first obtain the NF of the cascade using a direct method; according to (2.115), we simply calculate the total noise at the output due to the two stages, divide by $(V_{out}/V_{in})^2$, normalize to $4kTR_S$, and add one to the result. Taking the loadings into account, we write the overall voltage gain as

$$A_0 = \frac{V_{out}}{V_{in}} = \frac{R_{in1}}{R_{in1} + R_S} A_{v1} \frac{R_{in2}}{R_{in2} + R_{out1}} A_{v2}. \tag{2.123}$$

19. We assume for simplicity that the reactive components of the input and output impedances are nulled but the final result is valid even if they are not.

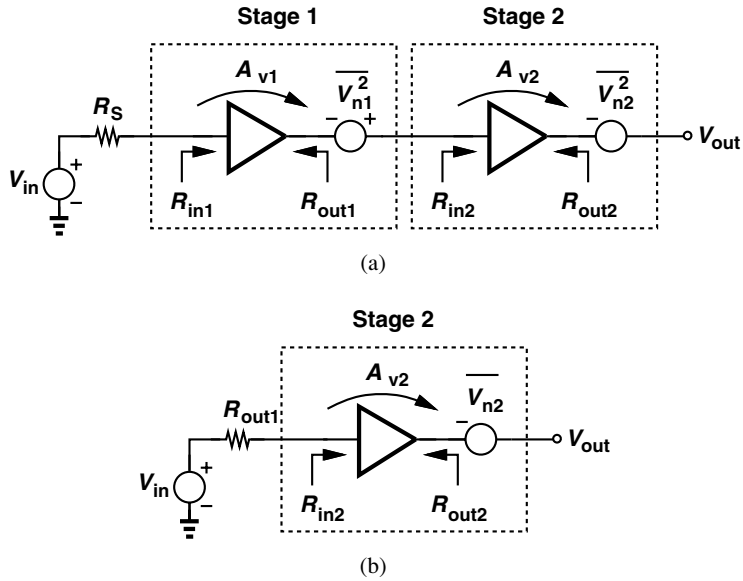


Figure 2.51 *(a) Noise in a cascade of stages, (b) simplified diagram.*

The output noise due to the two stages, denoted by $\overline{V_{n,out}^2}$, consists of two components: (a) $\overline{V_{n2}^2}$, and (b) $\overline{V_{n1}^2}$ amplified by the second stage. Since V_{n1} sees an impedance of R_{out1} to its left and R_{in2} to its right, it is scaled by a factor of $R_{in2}/(R_{in2} + R_{out1})$ as it appears at the input of the second stage. Thus,

$$\overline{V_{n,out}^2} = \overline{V_{n2}^2} + \overline{V_{n1}^2}\frac{R_{in2}^2}{(R_{in2} + R_{out1})^2}A_{v2}^2. \tag{2.124}$$

The overall NF is therefore expressed as

$$\text{NF}_{\text{tot}} = 1 + \frac{\overline{V_{n,out}^2}}{A_0^2} \cdot \frac{1}{4kTR_S} \tag{2.125}$$

$$= 1 + \frac{\overline{V_{n1}^2}}{\left(\dfrac{R_{in1}}{R_{in1} + R_S}\right)^2 A_{v1}^2} \cdot \frac{1}{4kTR_S} + \frac{\overline{V_{n2}^2}}{\left(\dfrac{R_{in1}}{R_{in1} + R_S}\right)^2 A_{v1}^2 \left(\dfrac{R_{in2}}{R_{in2} + R_{out1}}\right)^2 A_{v2}^2} \cdot \frac{1}{4kTR_S} \tag{2.126}$$

The first two terms constitute the *NF* of the first stage, NF_1, with respect to a source impedance of R_S. The third term represents the noise of the second stage, but how can it be expressed in terms of the *noise figure* of this stage?

Let us now consider the second stage by itself and determine its noise figure with respect to a source impedance of R_{out1} [Fig. 2.51(b)]. Using (2.115) again, we have

$$NF_2 = 1 + \frac{\overline{V_{n2}^2}}{\dfrac{R_{in2}^2}{(R_{in2}+R_{out1})^2}A_{v2}^2}\frac{1}{4kTR_{out1}}. \tag{2.127}$$

It follows from (2.126) and (2.127) that

$$NF_{tot} = NF_1 + \frac{NF_2 - 1}{\dfrac{R_{in1}^2}{(R_{in1}+R_S)^2}A_{v1}^2\dfrac{R_S}{R_{out1}}}. \tag{2.128}$$

What does the denominator represent? This quantity is in fact the "available power gain" of the first stage, defined as the "available power" at its output, $P_{out,av}$ (the power that it would deliver to a matched load) divided by the available source power, $P_{S,av}$ (the power that the source would deliver to a matched load). This can be readily verified by finding the power that the first stage in Fig. 2.51(a) would deliver to a load equal to R_{out1}:

$$P_{out,av} = V_{in}^2\frac{R_{in1}^2}{(R_S+R_{in1})^2}A_{v1}^2 \cdot \frac{1}{4R_{out1}}. \tag{2.129}$$

Similarly, the power that V_{in} would deliver to a load of R_S is given by

$$P_{S,av} = \frac{V_{in}^2}{4R_S}. \tag{2.130}$$

The ratio of (2.129) and (2.130) is indeed equal to the denominator in (2.128).

With these observations, we write

$$NF_{tot} = NF_1 + \frac{NF_2 - 1}{A_{P1}}, \tag{2.131}$$

where A_{P1} denotes the "available power gain" of the first stage. It is important to bear in mind that NF_2 is computed with respect to the output impedance of the first stage. For m stages,

$$NF_{tot} = 1 + (NF_1 - 1) + \frac{NF_2 - 1}{A_{P1}} + \cdots + \frac{NF_m - 1}{A_{P1}\cdots A_{P(m-1)}}. \tag{2.132}$$

Called "Friis' equation" [7], this result suggests that the noise contributed by each stage decreases as the total gain preceding that stage increases, implying that the first few stages in a cascade are the most critical. Conversely, if a stage suffers from attenuation (loss), then the NF of the following circuits is "amplified" when referred to the input of that stage.

Example 2.22

Determine the *NF* of the cascade of common-source stages shown in Fig. 2.52. Neglect the transistor capacitances and flicker noise.

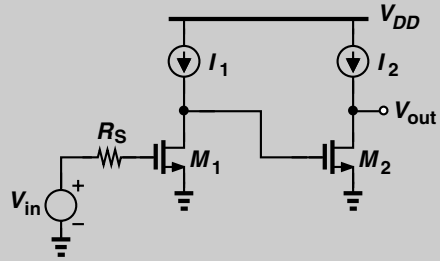


Figure 2.52 *Cascade of CS stages for noise figure calculation.*

Solution:

Which approach is simpler to use here, the direct method or Friis' equation? Since $R_{in1} = R_{in2} = \infty$, Eq. (2.126) reduces to

$$NF = 1 + \frac{\overline{V_{n1}^2}}{A_{v1}^2}\frac{1}{4kTR_S} + \frac{\overline{V_{n2}^2}}{A_{v1}^2 A_{v2}^2}\frac{1}{4kTR_S}, \tag{2.133}$$

where $\overline{V_{n1}^2} = 4kT\gamma g_{m1} r_{O1}^2$, $\overline{V_{n2}^2} = 4kT\gamma g_{m2} r_{O2}^2$, $A_{v1} = g_{m1} r_{O1}$, and $A_{v2} = g_{m2} r_{O2}$. With all of these quantities readily available, we simply substitute for their values in (2.133), obtaining

$$NF = 1 + \frac{\gamma}{g_{m1} R_S} + \frac{\gamma}{g_{m1}^2 r_{O1}^2 g_{m2} R_S}. \tag{2.134}$$

On the other hand, Friis' equation requires the calculation of the available power gain of the first stage and the *NF* of the second stage with respect to a source impedance of r_{O1}, leading to lengthy algebra.

The foregoing example represents a typical situation in modern RF design: the interface between the two stages does not have a 50-Ω impedance *and* no attempt has been made to provide impedance matching between the two stages. In such cases, Friis' equation becomes cumbersome, making direct calculation of the *NF* more attractive.

While the above example assumes an infinite input impedance for the second stage, the direct method can be extended to more realistic cases with the aid of Eq. (2.126). Even in the presence of complex input and output impedances, Eq. (2.126) indicates that (1) $\overline{V_{n1}^2}$ must be divided by the *unloaded* gain from V_{in} to the output of the first stage; (2) the output noise of the second stage, $\overline{V_{n2}^2}$, must be calculated with this stage driven by the output impedance of the first stage;[20] and (3) $\overline{V_{n2}^2}$ must be divided by the total voltage gain from V_{in} to V_{out}.

20. Recall from Example 2.19 that the output noise of a circuit may depend on the source impedance driving it, but the source impedance noise is excluded from $\overline{V_{n2}^2}$.

Example 2.23

Determine the noise figure of the circuit shown in Fig. 2.53(a). Neglect transistor capacitances, flicker noise, channel-length modulation, and body effect.

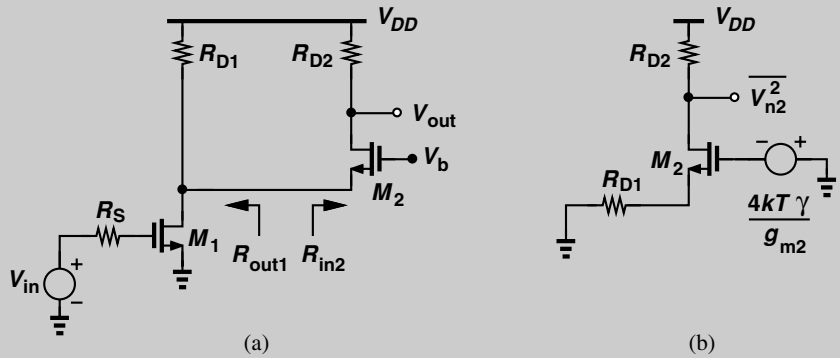


Figure 2.53 *(a) Cascade of CS and CG stages, (b) simplified diagram.*

Solution:

For the first stage, $A_{v1} = -g_{m1}R_{D1}$ and the unloaded output noise is equal to

$$\overline{V_{n1}^2} = 4kT\gamma g_{m1}R_{D1}^2 + 4kTR_{D1}. \tag{2.135}$$

For the second stage, the reader can show from Fig. 2.53(b) that

$$\overline{V_{n2}^2} = \frac{4kT\gamma}{g_{m2}}\left(\frac{R_{D2}}{\frac{1}{g_{m2}} + R_{D1}}\right)^2 + 4kTR_{D2}. \tag{2.136}$$

Note that the output impedance of the first stage is included in the calculation of $\overline{V_{n2}^2}$ but the *noise* of R_{D1} is not.

We now substitute these values in Eq. (2.126), bearing in mind that $R_{in2} = 1/g_{m2}$ and $A_{v2} = g_{m2}R_{D2}$.

$$NF_{tot} = 1 + \frac{4kT\gamma g_{m1}R_D^2 + 4kTR_{D1}}{g_{m1}^2R_{D1}^2}\cdot\frac{1}{4kTR_S} + \frac{\dfrac{4kT\gamma}{g_{m2}}\left(\dfrac{R_{D2}}{g_{m2}^{-1} + R_{D2}}\right)^2 + 4kTR_{D2}}{g_{m1}^2R_{D1}^2\left(\dfrac{g_{m2}^{-1}}{g_{m2}^{-1} + R_{D1}}\right)^2 g_{m2}^2R_{D2}^2}\cdot\frac{1}{4kTR_S}. \tag{2.137}$$

Noise Figure of Lossy Circuits Passive circuits such as filters appear at the front end of RF transceivers and their loss proves critical (Chapter 4). The loss arises from unwanted

resistive components within the circuit that convert the input power to heat, thereby producing a smaller signal power at the output. Furthermore, recall from Fig. 2.37 that resistive components also *generate* thermal noise. That is, passive lossy circuits both attenuate the signal and introduce noise.

We wish to prove that the noise figure of a passive (reciprocal) circuit is equal to its "power loss," defined as $L = P_{in}/P_{out}$, where P_{in} is the available source power and P_{out} the available power at the output. As mentioned in the derivation of Friis' equation, the available power is the power that a given source or circuit would deliver to a conjugate-matched load. The proof is straightforward if the input and output are matched (Problem 2.20). We consider a more general case here.

Consider the arrangement shown in Fig. 2.54(a), where the lossy circuit is driven by a source impedance of R_S while driving a load impedance of R_L.[21] From Eq. (2.130), the available source power is $P_{in} = V_{in}^2/(4R_S)$. To determine the available output power, we construct the Thevenin equivalent shown in Fig. 2.54(b), obtaining $P_{out} = V_{Thev}^2/(4R_{out})$. Thus, the loss is given by

$$L = \frac{V_{in}^2}{V_{Thev}^2}\frac{R_{out}}{R_S}. \tag{2.138}$$

To calculate the noise figure, we utilize the theorem illustrated in Fig. 2.37 and the equivalent circuit shown in Fig. 2.54(c) to write

$$\overline{V_{n,out}^2} = 4kTR_{out}\frac{R_L^2}{(R_L + R_{out})^2}. \tag{2.139}$$

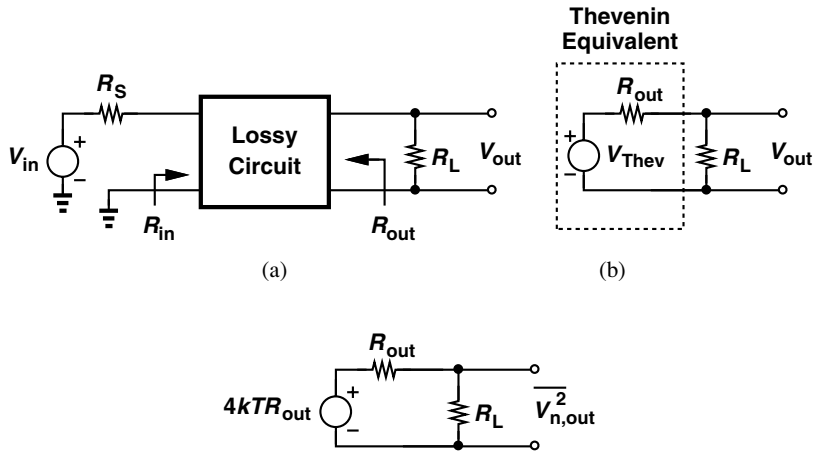


Figure 2.54 *(a) Lossy passive network, (b) Thevenin equivalent, (c) simplified diagram.*

21. For simplicity, we assume the reactive parts of the impedances are cancelled but the final result is valid even if they are not.

Note that R_L is assumed noiseless so that only the noise figure of the lossy circuit can be determined. The voltage gain from V_{in} to V_{out} is found by noting that, in response to V_{in}, the circuit produces an output voltage of $V_{out} = V_{Thev}R_L/(R_L + R_{out})$ [Fig. 2.54(b)]. That is,

$$A_0 = \frac{V_{Thev}}{V_{in}} \frac{R_L}{R_L + R_{out}}. \tag{2.140}$$

The NF is equal to (2.139) divided by the square of (2.140) and normalized to $4kTR_S$:

$$NF = 4kTR_{out} \frac{V_{in}^2}{V_{Thev}^2} \frac{1}{4kTR_S} \tag{2.141}$$

$$= L. \tag{2.142}$$

Example 2.24

The receiver shown in Fig. 2.55 incorporates a front-end band-pass filter (BPF) to suppress some of the interferers that may desensitize the LNA. If the filter has a loss of L and the LNA a noise figure of NF_{LNA}, calculate the overall noise figure.

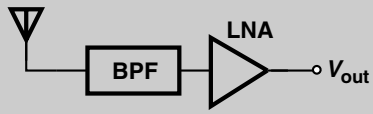


Figure 2.55 *Cascade of BPF and LNA.*

Solution:

Denoting the noise figure of the filter by NF_{filt}, we write Friis' equation as

$$NF_{tot} = NF_{filt} + \frac{NF_{LNA} - 1}{L^{-1}} \tag{2.143}$$

$$= L + (NF_{LNA} - 1)L \tag{2.144}$$

$$= L \cdot NF_{LNA}, \tag{2.145}$$

where NF_{LNA} is calculated with respect to the output resistance of the filter. For example, if $L = 1.5$ dB and $NF_{LNA} = 2$ dB, then $NF_{tot} = 3.5$ dB.

2.4 SENSITIVITY AND DYNAMIC RANGE

The performance of RF receivers is characterized by many parameters. We study two, namely, sensitivity and dynamic range, here and defer the others to Chapter 3.

2.4.1 Sensitivity

The sensitivity is defined as the minimum signal level that a receiver can detect with "acceptable quality." In the presence of excessive noise, the detected signal becomes unintelligible and carries little information. We define acceptable quality as sufficient signal-to-noise ratio, which itself depends on the type of modulation and the corruption (e.g., bit error rate) that the system can tolerate. Typical required SNR levels are in the range of 6 to 25 dB (Chapter 3).

In order to calculate the sensitivity, we write

$$NF = \frac{SNR_{in}}{SNR_{out}} \tag{2.146}$$

$$= \frac{P_{sig}/P_{RS}}{SNR_{out}}, \tag{2.147}$$

where P_{sig} denotes the input signal power and P_{RS} the source resistance noise power, both per unit bandwidth. Do we express these quantities in V^2/Hz or W/Hz? Since the input impedance of the receiver is typically matched to that of the antenna (Chapter 4), the antenna indeed delivers signal power and noise power to the receiver. For this reason, it is common to express both quantities in W/Hz (or dBm/Hz). It follows that

$$P_{sig} = P_{RS} \cdot NF \cdot SNR_{out}. \tag{2.148}$$

Since the overall signal power is distributed across a certain bandwidth, B, the two sides of (2.148) must be integrated over the bandwidth so as to obtain the total mean squared power. Assuming a flat spectrum for the signal and the noise, we have

$$P_{sig,tot} = P_{RS} \cdot NF \cdot SNR_{out} \cdot B. \tag{2.149}$$

Equation (2.149) expresses the sensitivity as the minimum input signal that yields a given value for the output *SNR*. Changing the notation slightly and expressing the quantities in dB or dBm, we have[22]

$$P_{sen}|_{dBm} = P_{RS}|_{dBm/Hz} + NF|_{dB} + SNR_{min}|_{dB} + 10\log B, \tag{2.150}$$

where P_{sen} is the sensitivity and B is expressed in Hz. Note that (2.150) does not directly depend on the gain of the system. If the receiver is matched to the antenna, then from (2.91), $P_{RS} = kT = -174$ dBm/Hz and

$$P_{sen} = -174\,\text{dBm/Hz} + NF + 10\log B + SNR_{min}. \tag{2.151}$$

Note that the sum of the first three terms is the total integrated noise of the system (sometimes called the "noise floor").

22. Note that in conversion to dB or dBm, we take 10 log because these are power quantities.

Example 2.25

A GSM receiver requires a minimum *SNR* of 12 dB and has a channel bandwidth of 200 kHz. A wireless LAN receiver, on the other hand, specifies a minimum *SNR* of 23 dB and has a channel bandwidth of 20 MHz. Compare the sensitivities of these two systems if both have an *NF* of 7 dB.

Solution:

For the GSM receiver, $P_{sen} = -102$ dBm, whereas for the wireless LAN system, $P_{sen} = -71$ dBm. Does this mean that the latter is inferior? No, the latter employs a much wider bandwidth and a more efficient modulation to accommodate a data rate of 54 Mb/s. The GSM system handles a data rate of only 270 kb/s. In other words, specifying the sensitivity of a receiver without the data rate is not meaningful.

2.4.2 Dynamic Range

Dynamic range (DR) is loosely defined as the maximum input level that a receiver can "tolerate" divided by the minimum input level that it can detect (the sensitivity). This definition is quantified differently in different applications. For example, in analog circuits such as analog-to-digital converters, the DR is defined as the "full-scale" input level divided by the input level at which $SNR = 1$. The full scale is typically the input level beyond which a hard saturation occurs and can be easily determined by examining the circuit.

In RF design, on the other hand, the situation is more complicated. Consider a simple common-source stage. How do we define the input "full scale" for such a circuit? Is there a particular input level beyond which the circuit becomes excessively nonlinear? We may view the 1-dB compression point as such a level. But, what if the circuit senses two interferers and suffers from intermodulation?

In RF design, two definitions of DR have emerged. The first, simply called the dynamic range, refers to the maximum tolerable *desired* signal power divided by the minimum tolerable desired signal power (the sensitivity). Illustrated in Fig. 2.56(a), this DR is limited by compression at the upper end and noise at the lower end. For example, a cell phone coming close to a base station may receive a very large signal and must process it with

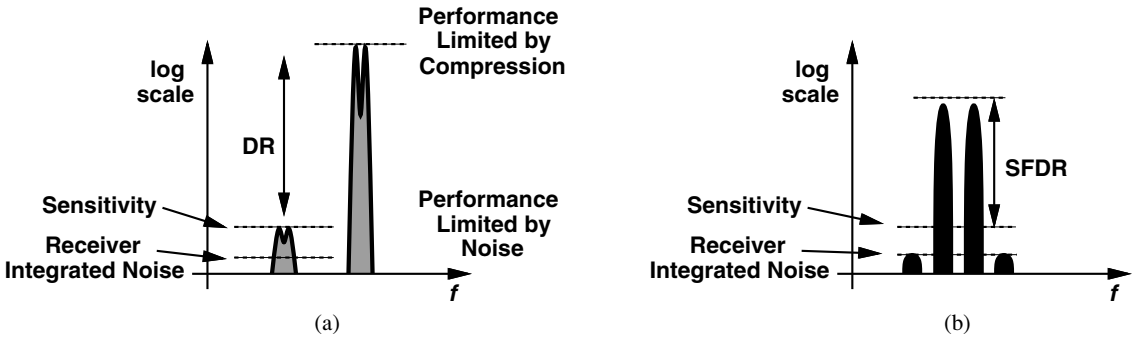


Figure 2.56 *Definitions of (a) DR and (b) SFDR.*

acceptable distortion. In fact, the cell phone measures the signal strength and adjusts the receiver gain so as to avoid compression. Excluding interferers, this "compression-based" DR can exceed 100 dB because the upper end can be raised relatively easily.

The second type, called the "spurious-free dynamic range" (SFDR), represents limitations arising from both noise and interference. The lower end is still equal to the sensitivity, but the upper end is defined as the maximum input level in a *two-tone* test for which the third-order IM products do not exceed the integrated noise of the receiver. As shown in Fig. 2.56(b), two (modulated or unmodulated) tones having equal amplitudes are applied and their level is raised until the IM products reach the integrated noise.[23] The ratio of the power of each tone to the sensitivity yields the SFDR. The SFDR represents the maximum relative level of interferers that a receiver can tolerate while producing an acceptable signal quality from a small input level.

Where should the various levels depicted in Fig. 2.56(b) be measured, at the input of the circuit or at its output? Since the IM components appear only at the output, the output port serves as a more natural candidate for such a measurement. In this case, the sensitivity—usually an input-referred quantity—must be scaled by the gain of the circuit so that it is referred to the output. Alternatively, the output IM magnitudes can be divided by the gain so that they are referred to the input. We follow the latter approach in our SFDR calculations.

To determine the upper end of the SFDR, we rewrite Eq. (2.56) as

$$P_{IIP3} = P_{in} + \frac{P_{out} - P_{IM,out}}{2}, \tag{2.152}$$

where, for the sake of brevity, we have denoted $20\log A_x$ as P_x even though no actual power may be transferred at the input or output ports. Also, $P_{IM,out}$ represents the level of IM products at the output. If the circuit exhibits a gain of G (in dB), then we can refer the IM level to the input by writing $P_{IM,in} = P_{IM,out} - G$. Similarly, the *input* level of each tone is given by $P_{in} = P_{out} - G$. Thus, (2.152) reduces to

$$P_{IIP3} = P_{in} + \frac{P_{in} - P_{IM,in}}{2} \tag{2.153}$$

$$= \frac{3P_{in} - P_{IM,in}}{2}, \tag{2.154}$$

and hence

$$P_{in} = \frac{2P_{IIP3} + P_{IM,in}}{3}. \tag{2.155}$$

The upper end of the SFDR is that value of P_{in} which makes $P_{IM,in}$ equal to the integrated noise of the receiver:

$$P_{in,max} = \frac{2P_{IIP3} + (-174\,\text{dBm} + NF + 10\log B)}{3}. \tag{2.156}$$

23. Note that the integrated noise is a single value (e.g., 100 μV_{rms}), not a *density*.

The SFDR is the difference (in dB) between $P_{in,max}$ and the sensitivity:

$$SFDR = P_{in,max} - (-174\,\text{dBm} + NF + 10\log B + SNR_{min}) \tag{2.157}$$

$$= \frac{2(P_{IIP3} + 174\,\text{dBm} - NF - 10\log B)}{3} - SNR_{min}. \tag{2.158}$$

For example, a GSM receiver with $NF = 7$ dB, $P_{IIP3} = -15$ dBm, and $SNR_{min} = 12$ dB achieves an SFDR of 54 dB, a substantially lower value than the dynamic range in the absence of interferers.

Example 2.26

The upper end of the dynamic range is limited by intermodulation in the presence of *two* interferers or desensitization in the presence of *one* interferer. Compare these two cases and determine which one is more restrictive.

Solution:

We must compare the upper end expressed by Eq. (2.156) with the 1-dB compression point:

$$P_{1-dB} \overset{?}{\gtrless} P_{in,max}. \tag{2.159}$$

Since $P_{1-dB} = P_{IIP3} - 9.6$ dB,

$$P_{IIP3} - 9.6\,\text{dB} \overset{?}{\gtrless} \frac{2P_{IIP3} + (-174\,\text{dBm} + NF + 10\log B)}{3} \tag{2.160}$$

and hence

$$P_{IIP3} - 28.8\,\text{dB} \overset{?}{\gtrless} -174\,\text{dBm} + NF + 10\log B. \tag{2.161}$$

Since the right-hand side represents the receiver noise floor, we expect it to be much lower than the left-hand side. In fact, even for an extremely wideband channel of $B = 1$ GHz and $NF = 10$ dB, the right-hand side is equal to -74 dBm, whereas, with a typical P_{IIP3} of -10 to -25 dBm, the left-hand side still remains higher. It is therefore plausible to conclude that

$$P_{1-dB} > P_{in,max}. \tag{2.162}$$

It follows that the maximum tolerable level in a two-tone test is quite lower than that in a compression test, i.e., corruption by intermodulation between two interferers is much greater than compression due to one. The SFDR is therefore a more stringent characteristic of the system than the compression-based dynamic range.

2.5 PASSIVE IMPEDANCE TRANSFORMATION

At radio frequencies, we often employ passive networks to transform impedances—from high to low and vice versa, or from complex to real and vice versa. Called "matching

networks," such circuits do not easily lend themselves to integration because their constituent devices, particularly inductors, suffer from loss if built on silicon chips. (We do use on-chip inductors in many RF building blocks.) Nonetheless, a basic understanding of impedance transformation is essential.

2.5.1 Quality Factor

In its simplest form, the quality factor, Q, indicates how close to ideal an energy-storing device is. An ideal capacitor dissipates no energy, exhibiting an infinite Q, but a series resistance, R_S [Fig. 2.57(a)], reduces its Q to

$$Q_S = \frac{\frac{1}{C\omega}}{R_S}, \tag{2.163}$$

where the numerator denotes the "desired" component and the denominator, the "undesired" component. If the resistive loss in the capacitor is modeled by a *parallel* resistance [Fig. 2.57(b)], then we must define the Q as

$$Q_P = \frac{R_P}{\frac{1}{C\omega}}, \tag{2.164}$$

because an ideal (infinite Q) results only if $R_P = \infty$. As depicted in Figs. 2.57(c) and (d), similar concepts apply to inductors

$$Q_S = \frac{L\omega}{R_S} \tag{2.165}$$

$$Q_P = \frac{R_P}{L\omega}. \tag{2.166}$$

While a parallel resistance appears to have no physical meaning, modeling the loss by R_P proves useful in many circuits such as amplifiers and oscillators (Chapters 5 and 8). We will also introduce other definitions of Q in Chapter 8.

2.5.2 Series-to-Parallel Conversion

Before studying transformation techniques, let us consider the series and parallel *RC* sections shown in Fig. 2.58. What choice of values makes the two networks equivalent?

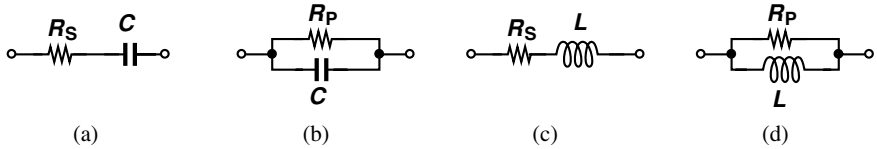


Figure 2.57 *(a) Series RC circuit, (b) equivalent parallel circuit, (c) series RL circuit, (d) equivalent parallel circuit.*

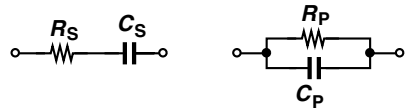


Figure 2.58 *Series-to-parallel conversion.*

Equating the impedances,

$$\frac{R_S C_S s + 1}{C_S s} = \frac{R_P}{R_P C_P s + 1}, \tag{2.167}$$

and substituting $j\omega$ for s, we have

$$R_P C_S j\omega = 1 - R_P C_P R_S C_S \omega^2 + (R_P C_P + R_S C_S) j\omega, \tag{2.168}$$

and hence

$$R_P C_P R_S C_S \omega^2 = 1 \tag{2.169}$$

$$R_P C_P + R_S C_S - R_P C_S = 0. \tag{2.170}$$

Equation (2.169) implies that $Q_S = Q_P$.

Of course, the two impedances cannot remain equal at all frequencies. For example, the series section approaches an open circuit at low frequencies while the parallel section does not. Nevertheless, an approximation allows equivalence for a narrow frequency range. We first substitute for $R_P C_P$ in (2.169) from (2.170), obtaining

$$R_P = \frac{1}{R_S C_S^2 \omega^2} + R_S. \tag{2.171}$$

Utilizing the definition of Q_S in (2.163), we have

$$R_P = (Q_S^2 + 1) R_S. \tag{2.172}$$

Substitution in (2.169) thus yields

$$C_P = \frac{Q_S^2}{Q_S^2 + 1} C_S. \tag{2.173}$$

So long as $Q_S^2 \gg 1$ (which is true for a finite frequency range),

$$R_P \approx Q_S^2 R_S \tag{2.174}$$

$$C_P \approx C_S. \tag{2.175}$$

That is, the series-to-parallel conversion retains the value of the capacitor but raises the resistance by a factor of Q_S^2. These approximations for R_P and C_P are relatively accurate because the quality factors encountered in practice typically exceed 4. Conversely,

parallel-to-series conversion reduces the resistance by a factor of Q_P^2. This statement applies to *RL* sections as well.

2.5.3 Basic Matching Networks

A common situation in RF transmitter design is that a load resistance must be transformed to a lower value. The circuit shown in Fig. 2.59(a) accomplishes this task. As mentioned above, the capacitor in parallel with R_L converts this resistance to a lower *series* component [Fig. 2.59(b)]. The inductance is inserted to cancel the equivalent series capacitance.

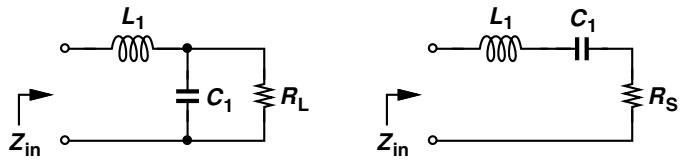


Figure 2.59 *(a) Matching network, (b) equivalent circuit.*

Writing Z_{in} from Fig. 2.59(a) and replacing s with $j\omega$, we have

$$Z_{in}(j\omega) = \frac{R_L(1 - L_1C_1\omega^2) + jL_1\omega}{1 + jR_LC_1\omega}. \tag{2.176}$$

Thus,

$$Re\{Z_{in}\} = \frac{R_L}{1 + R_L^2C_1^2\omega^2} \tag{2.177}$$

$$= \frac{R_L}{1 + Q_P^2}, \tag{2.178}$$

indicating that R_L is transformed down by a factor of $1 + Q_P^2$. Also, setting the imaginary part to zero gives

$$L_1 = \frac{R_L^2C_1}{1 + R_L^2C_1^2\omega^2} \tag{2.179}$$

$$= \frac{R_L^2C_1}{1 + Q_P^2}. \tag{2.180}$$

If $Q_P^2 \gg 1$, then

$$Re\{Z_{in}\} \approx \frac{1}{R_LC_1^2\omega^2} \tag{2.181}$$

$$L_1 = \frac{1}{C_1\omega^2}. \tag{2.182}$$

The following example illustrates how the component values are chosen.

Example 2.27

Design the matching network of Fig. 2.59(a) so as to transform $R_L = 50\ \Omega$ to 25 Ω at a center frequency of 5 GHz.

Solution:

Assuming $Q_P^2 \gg 1$, we have from Eqs. (2.181) and (2.182), $C_1 = 0.90\,\text{pF}$ and $L_1 = 1.13\,\text{nH}$, respectively. Unfortunately, however, $Q_P = 1.41$, indicating that Eqs. (2.178) and (2.180) must be used instead. We thus obtain $C_1 = 0.637\,\text{pF}$ and $L_1 = 0.796\,\text{nH}$.

In order to transform a resistance to a higher value, the capacitive network shown in Fig. 2.60(a) can be used. The series-parallel conversion results derived previously provide insight here. If $Q^2 \gg 1$, the parallel combination of C_1 and R_L can be converted to a series network [Fig. 2.60(b)], where $R_S \approx [R_L(C_1\omega)^2]^{-1}$ and $C_S \approx C_1$. Viewing C_2 and C_1 as one capacitor, C_{eq}, and converting the resulting series section to a parallel circuit [Fig. 2.60(c)], we have

$$R_{tot} = \frac{1}{R_S(C_{eq}\omega)^2} \tag{2.183}$$

$$= \left(1 + \frac{C_1}{C_2}\right)^2 R_L. \tag{2.184}$$

That is, the network "boosts" the value of R_L by a factor of $(1 + C_1/C_2)^2$. Also,

$$C_{eq} = \frac{C_1 C_2}{C_1 + C_2}. \tag{2.185}$$

Note that the capacitive component must be cancelled by placing an inductor in *parallel* with the input.

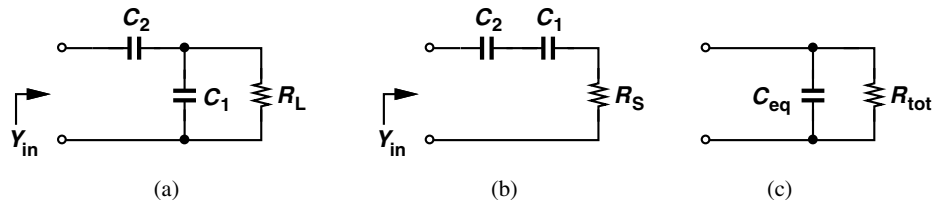


Figure 2.60 *(a) Capacitive matching circuit, (b) simplified circuit with parallel-to-series conversion, (c) simplified circuit with series-to-parallel conversion.*

For low Q values, the above derivations incur significant error. We thus compute the input admittance $(1/Y_{in})$ and replace s with $j\omega$,

$$Y_{in} = \frac{j\omega C_2(1 + j\omega R_L C_1)}{1 + R_L(C_1 + C_2)j\omega}. \tag{2.186}$$

The real part of Y_{in} yields the equivalent resistance seen to ground if we write

$$R_{tot} = \frac{1}{Re\{Y_{in}\}} \tag{2.187}$$

$$= \frac{1}{R_L C_2^2 \omega^2} + R_L \left(1 + \frac{C_1}{C_2}\right)^2. \tag{2.188}$$

In comparison with Eq. (2.184), this result contains an additional component, $(R_L C_2^2 \omega^2)^{-1}$.

Example 2.28

Determine how the circuit shown in Fig. 2.61(a) transforms R_L.

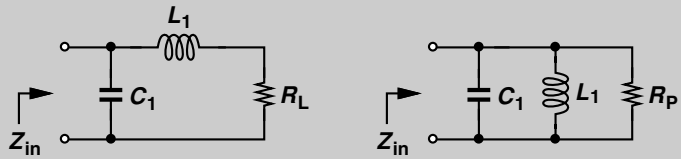


Figure 2.61 *(a) Matching network, (b) simplified circuit.*

Solution:

We postulate that conversion of the L_1–R_L branch to a parallel section produces a higher resistance. If $Q_S^2 = (L_1\omega/R_L)^2 \gg 1$, then the equivalent parallel resistance is obtained from Eq. (2.174) as

$$R_P = Q_S^2 R_L \tag{2.189}$$

$$= \frac{L_1^2 \omega^2}{R_L}. \tag{2.190}$$

The parallel equivalent inductance is approximately equal to L_1 and is cancelled by C_1 [Fig. 2.61(b)].

The intuition gained from our analysis of matching networks leads to the four "*L*-section" topologies[24] shown in Fig. 2.62. In Fig. 2.62(a), C_1 transforms R_L to a smaller series value and L_1 cancels C_1. Similarly, in Fig. 2.62(b), L_1 transforms R_L to a smaller series value while C_1 resonates with L_1. In Fig. 2.62(c), L_1 transforms R_L to a larger parallel value and C_1 cancels the resulting parallel inductance. A similar observation applies to Fig. 2.62(d).

How do these networks transform voltages and currents? As an example, consider the circuit in Fig. 2.62(a). For a sinusoidal input voltage with an rms value of V_{in}, the power

24. The term "*L*" is used because the capacitor and the inductor form the letter *L* in the circuit diagram.

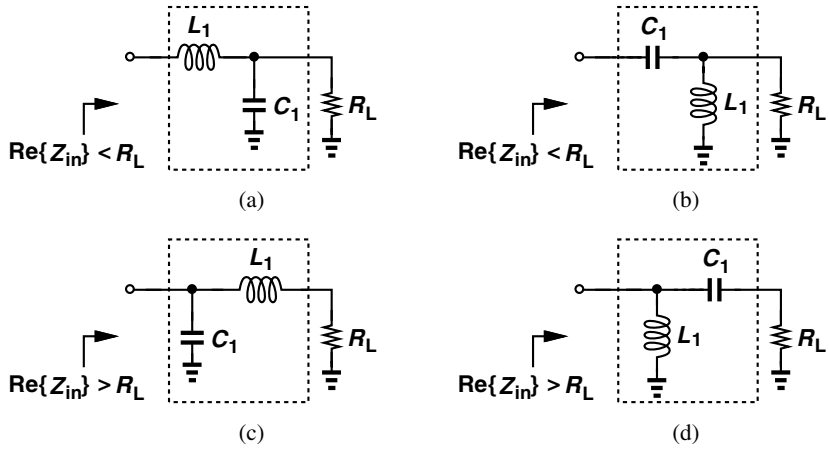


Figure 2.62 *Four L sections used for matching.*

delivered to the input port is equal to $V_{in}^2/Re\{Z_{in}\}$, and that delivered to the load, V_{out}^2/R_L. If L_1 and C_1 are ideal, these two powers must be equal, yielding

$$\frac{V_{out}}{V_{in}} = \sqrt{\frac{R_L}{Re\{Z_{in}\}}}. \tag{2.191}$$

This result, of course, applies to any lossless matching network whose input impedance contains a zero imaginary part. Since $P_{in} = V_{in}I_{in}$ and $P_{out} = V_{out}I_{out}$, we also have

$$\frac{I_{out}}{I_{in}} = \sqrt{\frac{Re\{Z_{in}\}}{R_L}}. \tag{2.192}$$

For example, a network transforming R_L to a *lower* value "amplifies" the voltage and attenuates the current by the above factor.

Example 2.29

A closer look at the *L*-sections in Figs. 2.62(a) and (c) suggests that one can be obtained from the other by swapping the input and output ports. Is it possible to generalize this observation?

Solution:

Yes, it is. Consider the arrangement shown in Fig. 2.63(a), where the passive network transforms R_L by a factor of α. Assuming the input port exhibits no imaginary component, we equate the power delivered to the network to the power delivered to the load:

$$\left(V_{in}\frac{\alpha R_L}{\alpha R_L + R_S}\right)^2 \cdot \frac{1}{\alpha R_L} = \frac{V_{out}^2}{R_L}. \tag{2.193}$$

Example 2.29 (*Continued*)

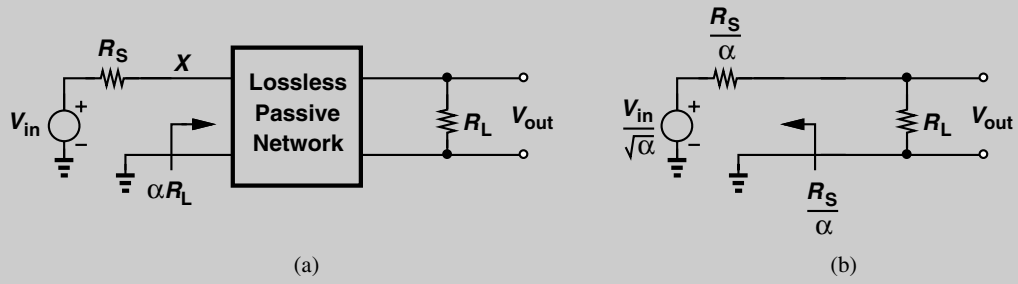


Figure 2.63 *(a) Input and (b) output impedances of a lossless passive network.*

It follows that

$$V_{out} = \frac{V_{in}}{\sqrt{\alpha}} \cdot \frac{R_L}{R_L + \dfrac{R_S}{\alpha}}, \tag{2.194}$$

pointing to the Thevenin equivalent shown in Fig. 2.63(b). We observe that the network transforms R_S by a factor of $1/\alpha$ and the input voltage by a factor of $1/\sqrt{\alpha}$, similar to that in Eq. (2.191). In other words, if the input and output ports of such a network are swapped, the resistance transformation ratio is simply inverted.

Transformers can also transform impedances. An ideal transformer having a turns ratio of n "amplifies" the input voltage by a factor of n (Fig. 2.64). Since no power is lost, $V_{in}^2/R_{in} = n^2 V_{in}^2/R_L$ and hence $R_{in} = R_L/n^2$. The behavior of actual transformers, especially those fabricated monolithically, is studied in Chapter 7.

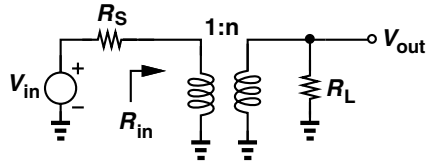


Figure 2.64 *Impedance transformation by a physical transformer.*

The networks studied here operate across only a narrow bandwidth because the transformation ratio, e.g., $1 + Q^2$, varies with frequency, and the capacitance and inductance approximately resonate over a narrow frequency range. Broadband matching networks can be constructed, but they typically suffer from a high loss.

2.5.4 Loss in Matching Networks

Our study of matching networks has thus far neglected the loss of their constituent components, particularly, that of inductors. We analyze the effect of loss in a few cases here, but, in general, simulations are necessary to determine the behavior of complex lossy networks.

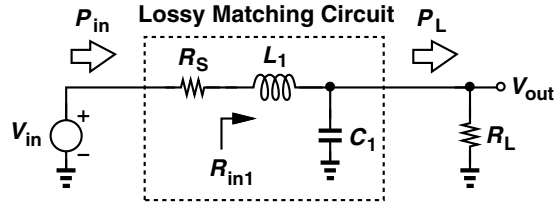


Figure 2.65 *Lossy matching network with series resistance.*

Consider the matching network of Fig. 2.62(a), shown in Fig. 2.65 with the loss of L_1 modeled by a series resistance, R_S. We define the loss as the power provided by the input divided by that delivered to R_L. The former is equal to

$$P_{in} = \frac{V_{in}^2}{R_S + R_{in1}} \tag{2.195}$$

and the latter,

$$P_L = \left(V_{in}\frac{R_{in1}}{R_S + R_{in1}}\right)^2 \cdot \frac{1}{R_{in1}}, \tag{2.196}$$

because the power delivered to R_{in1} is entirely absorbed by R_L. It follows that

$$\text{Loss} = \frac{P_{in}}{P_L} \tag{2.197}$$

$$= 1 + \frac{R_S}{R_{in1}}. \tag{2.198}$$

For example, if $R_S = 0.1R_{in1}$, then the (power) loss reaches 0.41 dB. Note that this network transforms R_L to a *lower* value, $R_{in1} = R_L/(1 + Q_P^2)$, thereby suffering from loss even if R_S appears small.

As another example, consider the network of Fig. 2.62(b), depicted in Fig. 2.66 with the loss of L_1 modeled by a parallel resistance, R_P. We note that the power delivered by V_{in}, P_{in}, is entirely absorbed by $R_P||R_L$:

$$P_{in} = \frac{V_{out}^2}{R_P||R_L} \tag{2.199}$$

$$= \frac{V_{out}^2}{R_L}\frac{R_P + R_L}{R_P}. \tag{2.200}$$

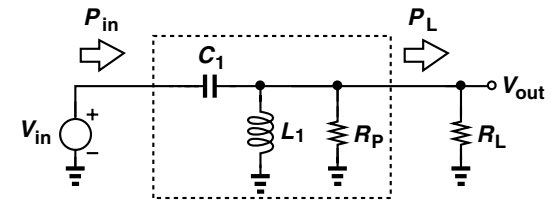


Figure 2.66 *Lossy matching network with parallel resistance.*